Test Booklet Code

ENGLISH **11**

NEET (UG)-2026

|| DATE: 03-05-2026 ||

Important Instructions:

1. The test is of **3 hours** duration and the Test Booklet contains **180** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry,** and **Biology (Botany and Zoology)**. All questions are compulsory.
2. Each question carries **4 marks**. For **each correct response**, the candidate will get **4 marks**. For **each incorrect response**, **1 mark will be deducted** from the total scores. The maximum mark is **720**.
3. Use a **Blue/Black** ballpoint Pen only for writing particulars on this page/marking responses on the Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy)** to the **Invigilator** before leaving the room/hall. The candidates are allowed to take away this Test Booklet with them.
6. The CODE for this Booklet is **11**.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet. The use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Admit Card to the Invigilator.
9. No candidate, without special permission of the Centre Superintendent or Invigilator, would leave his/her seat.
10. Use of an Electronic/Manual Calculator is prohibited.
11. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
12. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
13. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

1. The speed of light in vacuum is taken as unity. If light takes 6 min 40 s to reach the Earth from the Sun, the distance between the Sun and the Earth in new unit is:

(1) 3×10^{8}

(2) 500

(3) 3×10^{10}

(4) 400

2. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Young's Modulus	(I)	$\frac{\Delta d}{\Delta L}\left(\frac{L}{d}\right)$
(B)	Compressibility	(II)	$\frac{FL}{A(\Delta L)}$
(C)	Bulk Modulus	(III)	$-\frac{1}{\Delta P}\left(\frac{\Delta V}{V}\right)$
(D)	Poisson's Ratio	(IV)	$-P\left(\frac{V}{\Delta V}\right)$

Choose the **correct** answer from the options given below:

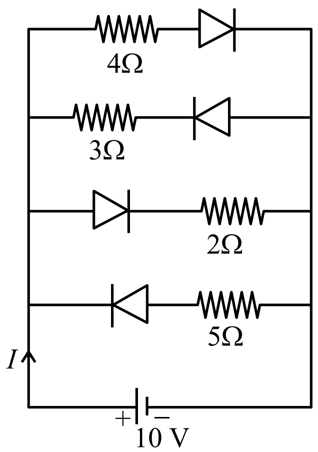
(1) A-IV, B-I, C-II, D-III

(2) A-III, B-II, C-I, D-IV

(3) A-I, B-IV, C-III, D-II

(4) A-II, B-III, C-IV, D-I

3. The current I in the circuit shown below is:
(All diodes are ideal and identical)



(1) $\frac{5}{3}$A

(2) $\frac{5}{9}$A

(3) $\frac{1}{3}$A

(4) $\frac{15}{2}$A

4. The angular speed of a flywheel is increased from 600 rpm to 1200 rpm in 10 s. The number of revolutions completed by the flywheel during this time is :

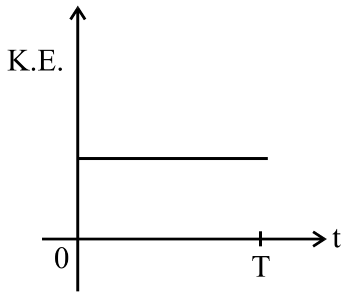
(1) 900

(2) 600

(3) 150

(4) 300

5. For a simple pendulum, having time period T, the variation of kinetic energy (K.E.) with time (t) is represented by:

(1)


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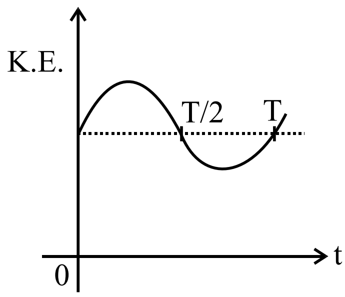

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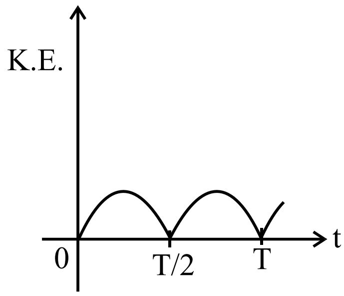

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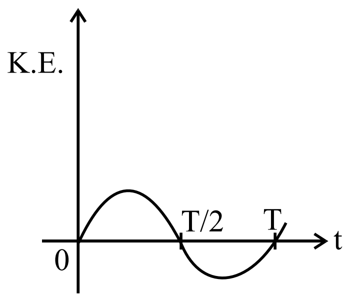


6. A resistor is connected to a battery of 12 V emf and internal resistance 2 Ω. If the current in the circuit is 0.6 A, the terminal voltage of the battery is:

(1) 10 V

(2) 1.2 V

(3) 12 V

(4) 10.8 V

7. A flask contains argon and chlorine in the ratio of 2 :1 by mass. The temperature of the mixture is 27°C. The ratio of root mean square speed of the molecules of the two gases $\left(\frac{v_{rms}^{Ar}}{v_{rms}^{Cl}}\right)$ is :

(Atomic mass of argon = 40.0 u and molecular mass of chlorine = 70.0 u)

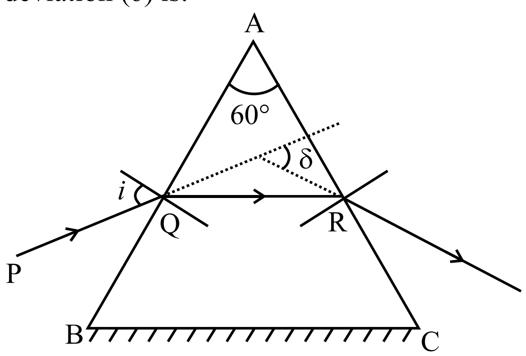
(1) $\frac{\sqrt{7}}{2}$

(2) $\frac{7}{4}$

(3) $\frac{7}{2}$

(4) $\frac{2}{\sqrt{7}}$

8. A ray of monochromatic light is passing through an equilateral prism (ABC) as shown in the figure. The refracted ray (QR) is parallel to its base (BC) and the angle of incidence (i) is 50°. Then the angle of deviation (δ) is:



(1) 45°
(2) 35°
(3) 40°
(4) 55°

9. Match **List-I** with **List-II**.

List-I		List-II	
(A)	$E = hv$	(I)	de Broglie wavelength
(B)	Diffraction and Interference	(II)	Particle nature of light
(C)	$\lambda = \frac{h}{p}$	(III)	Wave nature of light
(D)	Compton effect	(IV)	Energy of photon

Choose the **correct** answer from the options given below:

(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-IV, C-III, D-II
(4) A-IV, B-III, C-I, D-II

10. In the first excited state of hydrogen atom, the energy of its electron is –3.4 eV. The radial distance of the electron from the hydrogen nucleus in this case is approximately :

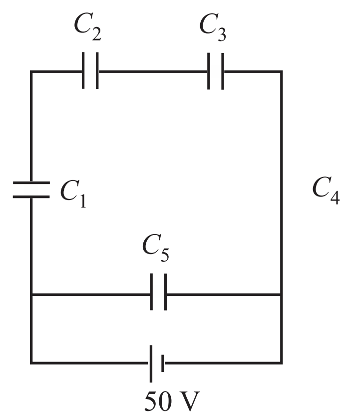
(Take 1 eV = 1.6×10^{-19} J, e = 1.6×10^{-19} C and $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ N m^2/C^2)

(1) 2.1×10^{-9} m (2) 2.1×10^{-8} m
(3) 2.1×10^{-10} m (4) 2.1×10^{-11} m

11. A box of mass 15 kg is kept on the floor of a stationary trolley. The coefficient of static friction between the box and the trolley is 0.12. Keeping the box in stationary state over the trolley, the maximum acceleration with which the trolley can be moved horizontally in m s^{-2} is :
(g = 10 m/s^2)

(1) 2.1 (2) 1.8
(3) 1.5 (4) 1.2

12. Five capacitors of capacitances $C_1 = C_2 = C_3 = C_4 = 10$ μF and $C_5 = 2.5$ μF are connected as shown, along with a battery of 50 V.



The equivalent capacitance and the charges on each capacitor respectively are:

(1) 5 μF, 125 μC on C_1 to C_4 and 25 μC on C_5
(2) 5 μF, 125 μC on all capacitors
(3) 5 μF, 250 μC on all capacitors
(4) 4 μF, 250 μC on C_1 to C_4 and 125 μC on C_5

13. The amount of work done to raise a mass 'm' from the surface of the Earth to a height equal to the radius of the Earth 'R', will be:

(1) 2 mg R

(2) mg$\frac{R}{4}$

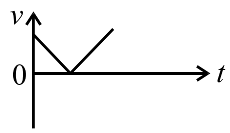
(3) mg R

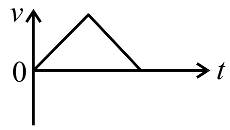
(4) mg$\frac{R}{2}$

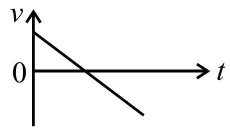
14. Each side of a metallic cube of mass 5.580 kg is measured to be 9.0 cm. Keeping the significant figures in view, the density of the material of the cube can be best expressed as $X \times 10^3$ kg m^{-3}, where the value of X is:

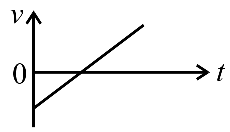
(1) 7.654 (2) 7.6
(3) 7.65 (4) 7.7

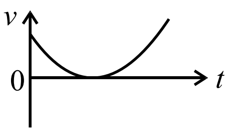
15 . The following plots show variation of velocity (v) with time (t), of a ball thrown vertically upward, and falling back. Which of the following plots is/are **correct**?

A. 

B. 

C. 

D. 

E. 

(1) C only (2) D only
(3) B only (4) A and E only

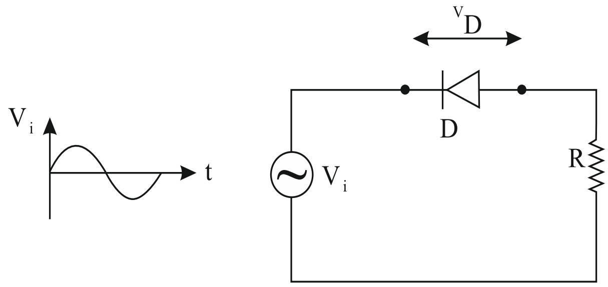
16. The sum of kinetic energy and potential energy of a simple pendulum bob is 0.02 joule. The speed of the simple pendulum bob at equilibrium position is approximately :
(Consider mass of the bob = 20 g)

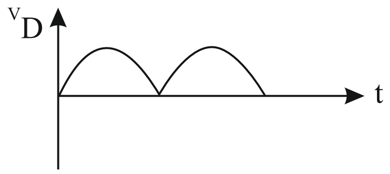
(1) 0.2 m/s (2) 1.41 m/s
(3) 14.1 m/s (4) 2.0 m/s

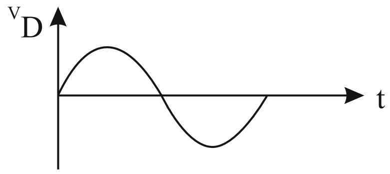
17. In Young's double slit experiment, using monochromatic light of wavelength λ, the intensity of light at a point on the screen where the path difference is λ is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{3}$ will be :

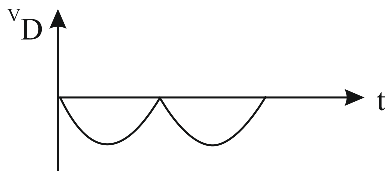
(1) $\frac{K}{4}$ (2) K
(3) 2 K (4) $\frac{K}{2}$

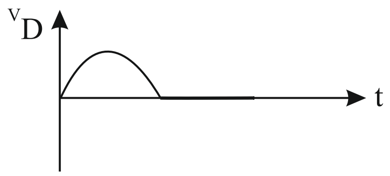
18. In the circuit shown below, the voltage appearing across the diode D will be of the form:



(1) 

(2) 

(3) 

(4) 

19. An ac circuit contains a resistance of $1\,k\Omega$, a capacitor of $0.1\,\mu F$ and an inductor of $1\,mH$ connected in series. The resonance frequency of the circuit is approximately:

(1) 13.5 kHz
(2) 10.1 kHz
(3) 20.7 kHz
(4) 15.9 kHz

20. In interference and diffraction, the light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing a bright fringe.

A. As there is no gain or loss of energy, these phenomena are consistent with the principle of conservation of energy.
B. Diffraction and interference are characteristics exhibited only by light waves.

Choose the correct answer from the options given below

(1) A is true and B is also true
(2) A is false, but B is true
(3) A is true, but B is false
(4) Both A and B are false

21. For a travelling harmonic wave
$y(x, t) = 2.0 \cos 2\pi(10t - 0.0080x + 0.35)$, where x and y are in cm and t in s. The phase difference between oscillatory motion of two points separated by a distance of 0.5 m is:
(1) 0.08π rad (2) 0.8π rad
(3) 8π rad (4) 0.008π rad

22. The magnitude and direction of the acceleration produced in a body of mass 5 kg when two mutually perpendicular forces 8 N and 6 N act on it, are respectively :
(1) $20\ m\ s^{-2}\ \tan^{-1}(4/3)$ with 8 N force
(2) $2\ m\ s^{-2} : \tan^{-1}(3/4)$ with 6 N force
(3) $2\ m\ s^{-2} ; \tan^{-1}(4/3)$ with 8 N force
(4) $2\ m\ s^{-2} ; \tan^{-1}(3/4)$ with 8 N force

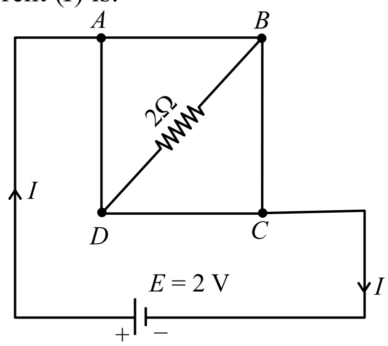
23. Consider two uncharged capacitors of equal capacitance 200 pF. One of them is charged by a 100 V supply and disconnected. Now this capacitor is connected to the uncharged capacitor. The amount of electrostatic energy lost in the process is:
(1) 0.5×10^{-6} J (2) 1.0 J
(3) 1.0×10^{-6} J (4) 0.5 J

24. The power of a crane, which lifts a mass of 1000 kg to a height of 20 m in 10 s is : $\left(g = 9.8\,m/s^2\right)$
(1) 19.6 W (2) 39.2 W
(3) 19.6 kW (4) 39.2 kW

25. In a vernier callipers, 20 VSD coincide with 16 MSD (each division of length 1 mm). The least count of the vernier callipers is:
(1) 0.2 cm
(2) 0.01 cm
(3) 0.02 cm
(4) 0.1 cm

26. When a ruler falls vertically, 5 different persons catch it with different reaction times.
$(g = 9.8\ ms^{-2})$
A. Person A has reaction time of 0.20 s.
B. Person B has reaction time of 0.22 s.
C. Person C has reaction time of 0.18 s.
D. Person D has reaction time of 0.19 s.
E. Person E has reaction time of 0.21 s.
What is the **correct** order of the distance travelled by the ruler for each person ?
(1) B > E > A > C > D
(2) C > D > A > B > E
(3) B > E > A > D > C
(4) C > D > A > E > B

27. A uniform metallic wire having resistance $4\,\Omega$ is bent to form a square loop (*ABCD*) (see figure). A resistance of $2\,\Omega$ is connected between points *B* and *D* and a battery of 2 V is connected across points *A* and *C* as shown in the figure. Now the value of current (*I*) is:



(1) 2 A (2) 8 A
(3) 4.5 A (4) 4 A

28. A room heater is rated 400 W, 220 V. If the supply voltage drops to 200 V, what will be the power consumed (approximately) ?
(1) 200 W (2) 400 W
(3) 331 W (4) 121 W

29. A 100-turn closely wound circular coil of radius 5 cm has a magnetic field of 3.14×10^{-3} T at its centre. The current flowing through the coil, and the magnitude of the magnetic moment of this coil are, respectively: (Take $\mu_0 = 4\pi \times 10^{-7}$ T m/A)
(1) $2\ A, 10\ A\ m^2$ (2) $2.5\ A, 20\ A\ m^2$
(3) $2\ A, 4\ A\ m^2$ (4) $2.5\ A, 2\ A\ m^2$

30. A rectangular wire loop of sides 8 cm and 3 cm with a small cut, is moving out of a region of uniform magnetic field of magnitude 0.3 T directed normal to the plane of the loop. The emf developed across the cut, if the velocity of the loop is 2 cm s^{-1}, in a direction normal to the shorter side of the loop, will be :
(1) 4.8×10^{-4} volt (2) 1.2×10^{-4} volt
(3) 1.3×10^{-4} volt (4) 1.8×10^{-4} volt

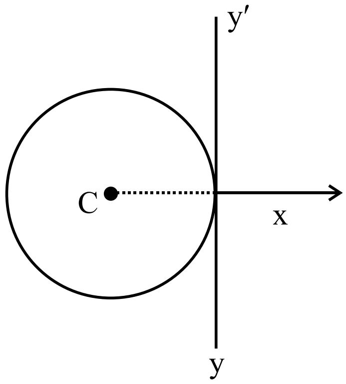
31. Four statements are given (A is mass number) :
A. The volume of a nucleus is proportional to $A^{1/3}$.
B. The volume of a nucleus is proportional to A.
C. The difference in mass of an atom and its nucleus is called the mass defect.
D. The difference in mass of a nucleus and its constituents is called the mass defect.
Choose the **correct** answer from the options given below :
(1) A and C are true, but B and D are false
(2) B and C are true, but A and D are false
(3) A and D are true, but B and C are false
(4) B and D are true, but A and C are false

32. An unknown nucleus has a nuclear density of 2.29×10^{17} kg/m^3 and mass of 19.926×10^{-27} kg. Its mass number A is approximately :
(Take $R_0 = 1.2 \times 10^{-15}$ m, $4\pi = 12.56$)
(1) 12
(2) 20
(3) 16
(4) 19

33. Savitha, a XI standard student, while conducting an experiment to determine the effective length of a simple pendulum L, notes down the data of time taken to complete 30 oscillations as 60 s and hence calculates the length of the simple pendulum as :
(Take $\pi^2 = 9.8$, and $g = 9.8\ m/s^2$)
(1) 0.75 m (2) 1.5 m
(3) 2 m (4) 1 m

34. An electric heater supplies heat to a system at a rate of 100 W. If the system performs work at a rate of 75 J/s, then the rate at which internal energy increases will be :
(1) 75 W (2) 100 W
(3) 125 W (4) 25 W

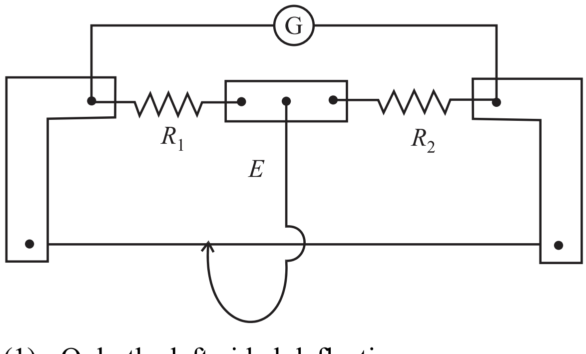
35. A thin wire of length 'L' and linear mass density 'm' is bent into a circular ring (in x − y plane) with centre 'C' as shown in figure. The moment of inertia of the ring about an axis yy' will be:



(1) $\frac{3\,mL^3}{8\pi}$ (2) $\frac{3\,mL^3}{8\pi^2}$
(3) $\frac{3\,mL^2}{8\pi}$ (4) $\frac{3\,mL^2}{8\pi^2}$

36. A galvanometer of resistance 100 Ω gives full scale deflection for a current of 1 mA. It is converted into an ammeter of range 0 – 10 A. The shunt required is:
(1) 0.01 Ω
(2) 0.10 Ω
(3) 1.0 Ω
(4) 0.001 Ω

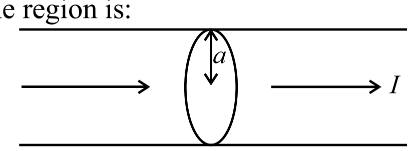
37. In a metre bridge experiment (see figure), the positions of the cell, E, and galvanometer, G, are interchanged. We shall observe in the galvanometer:

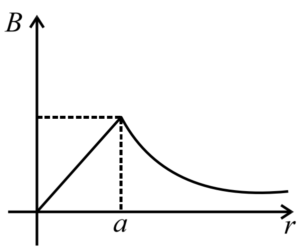


(1) Only the left-sided deflection
(2) There will be no deflection irrespective of the position of the jockey
(3) Only the right-sided deflection
(4) Both right-sided and left-sided deflection and at balance point, no-deflection

38. The peak value of an alternating current is 5 A and frequency is 60 Hz. How long will the current, starting from zero, take to reach the peak value ?
(1) $\frac{1}{120}$ s (2) $\frac{1}{60}$ s
(3) $\frac{1}{30}$ s (4) $\frac{1}{240}$ s

39. The figure given below shows a long straight solid wire of circular cross-section of radius 'a' carrying steady current I. The current I is uniformly distributed across its cross-section. The plot which correctly represents the variation of magnetic field (B) with distance (r) from the axis of the conductor in the region is:



(1)


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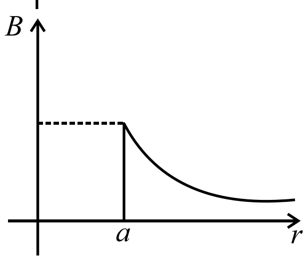
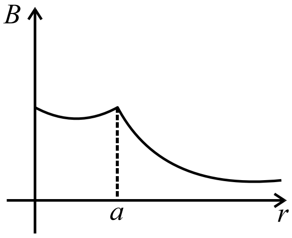
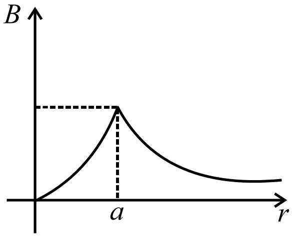


(3) 

(4) 

40. Two statements are given below :

A. When the forward bias voltage across a p-n junction diode increases above a certain threshold voltage, the diode current increases significantly.

B. This current is called reverse saturation current.

Choose the **correct** answer from the options given below :

(1) Both Statements A and B are false

(2) Statement A is true, but Statement B is false

(3) Both Statements A and B are true

(4) Statement A is false, but Statement B is true

41. Which of the following statements are correct ?

A. Inside a conductor, the electrostatic field is zero.

B. Electric field at the surface of a charged conductor does not depend on its surface charge density.

C. The interior of a charged conductor can have no excess charge in the static situation.

D. At the surface of a charged conductor, the electrostatic field must be normal to the surface at every point.

E. The electrostatic potential is zero everywhere inside a charged conductor.

Choose the **correct** answer from the options given below :

(1) A, B and D only

(2) A, C and E only

(3) A, C and D only

(4) C, D and E only

42. For a metal of work function 6.6 eV, which of the following wavelengths of incident radiation does ***not*** give rise to the photoelectric effect ?

(Take Planck's constant as 6.6×10^{-34} J s)

(1) 100 nm

(2) 150 nm

(3) 200 nm

(4) 50 nm

43. In a concave lens, a ray of light emanating from the object parallel to the principal axis of the lens, after refraction:

(1) passes through 2F, which is the radius of curvature of the lens.

(2) appears to diverge from the first principal focus.

(3) emerges parallel to the principal axis.

(4) passes through the second principal focus.

44. A submarine is designed to withstand an absolute pressure of 100 atm. How deep can it go below the water surface ?

(Consider the density of water = 1000 kg m^{-3}, 1 atm = 1×10^5 Pa and gravitational acceleration g = 10 m/s^2)

(1) 990 m

(2) 9900 m

(3) 99 m

(4) 9000 m

45. Match List I with List II :

List I (Electromagnetic wave)		List II (Production)	
A.	Microwave	I.	Electrons in atoms emit light when they move from a higher energy level to a lower energy level
B.	Visible light	II.	Radioactive decay of nucleus
C.	Gamma rays	III.	Vibration of atoms and molecules
D.	Infra-red rays	IV.	Klystron valve or magnetron valve

Choose the **correct** answer from the options given below :

(1) A-III, B-I, C-II, D-IV

(2) A-III, B-IV, C-I, D-II

(3) A-IV, B-I, C-II, D-III

(4) A-IV, B-III, C-II, D-I

46. Select the reagents that reduce nitriles to primary amines:

A. (i) $LiAlH_4$: (ii) H_2O
B. Sn + HCl
C. H_2/Ni
D. $Na(Hg)/C_2H_5OH$
E. Br_2/aq. NaOH

Choose the **correct** answer from the options given below:

(1) B, D and E only
(2) A, C and D only
(3) A, D and E only
(4) A, B and C only

47. Match **List-I** with **List-II**.

List-I (Transition metal/ compound/complex)		List-II (Catalytic Role)	
(A)	V_2O_5	(I)	Preparation of ammonia from N_2/H_2 mixture
(B)	Fe	(II)	Polymerisation of alkynes
(C)	$PdCl_2$	(III)	Preparation of H_2SO_4 from SO_2
(D)	Ni complex	(IV)	Oxidation of ethyne to ethanal

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-I, C-III, D-II
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-I, C-IV, D-II

48. Consider the following reaction:

$2A\ (g) + B\ (g) \rightarrow 2D\ (g)$

$\Delta U^{\ominus} = -10\ kJ\ mol^{-1}$ and $\Delta S^{\ominus} = -44\ J\ K^{-1}$ at 298 K.

Identify the **correct** option with $\Delta G^{\ominus}$ for the reaction and spontaneity of the reaction at 298 K.

(Given: $R = 8.31\ J\ mol^{-1}\ K^{-1}$)

(1) $-1.635\ kJ\ mol^{-1}$, spontaneous
(2) $-0.63568\ kJ\ mol^{-1}$, spontaneous
(3) $+0.63568\ kJ\ mol^{-1}$, non-spontaneous
(4) $+1.635\ kJ\ mol^{-1}$ non-spontaneous

49. Match **List-I** with **List-II**.

List-I (Quantum Numbers)			List-II (Orbital)	
	'n	'l'		
(A)	2	1	(I)	3d
(B)	4	0	(II)	2p
(C)	5	3	(III)	4s
(D)	3	2	(IV)	5f

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-III, D-I
(2) A-II, B-III, C-I, D-IV
(3) A-II, B-III, C-IV, D-I
(4) A-I, B-II, C-III, D-IV

50. In a qualitative analysis Bi^{3+} is detected by appearance of precipitate of BiO(OH)(s). Calculate pH when the following equilibrium exists at 298 K:

$BiO(OH)(s) \rightleftharpoons BiO^+(aq) + OH^-(aq)$,

$K = 4 \times 10^{-10}$

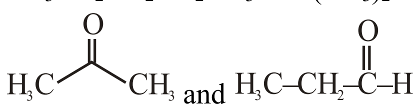
(Given: log 2 = 0.3010)

(1) 8.714
(2) 4.699
(3) 5.286
(4) 9.301

51. The **correct** statement with regard to the secondary structure of DNA/RNA is :

(1) RNA possesses a double strand helix structure and contains uracil as one of the four bases.
(2) DNA possesses a double strand helix structure and contains thymine as one of the four bases.
(3) RNA possesses a single strand helix structure and contains thymine as one of the four bases.
(4) DNA possesses a single strand helix structure and contains uracil as one of the four bases.

52. The pair of molecules that are metamers among the following is :

(1) $CH_3CH_2CH_2OH$ and $CH_3-CH(OH)-CH_3$
(2) $CH_3CH_2CH_2CH_2CH_3$ and $(CH_3)_2CHCH_2CH_3$
(3) 
(4) $CH_3OCH_2CH_2CH_3$ and $CH_3CH_2OCH_2CH_3$

53. Match List I with List II:

List I (Complex)	**List II (Type of isomerism)**
A. $[Pt(NH_3)_2Cl_2]$	I. Optical
B. $[Co(en)_3]^{3+}$	II. Solvate
C. $[Co(NH_3)_5]NO_2]Cl_2$	III. Geometrical
D. $[Cr(H_2O)_6]Cl_3$	IV. Linkage

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-II, D-IV
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-IV, C-III, D-I
(4) A-III, B-I, C-IV, D-II

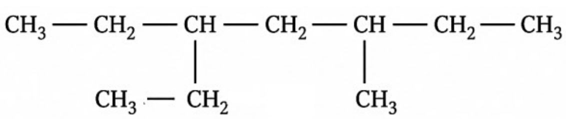
54. Match the **List-I** with **List-II**.

List-I (Order of reaction)		**List-II (Unit of rate constant)**	
(A)	Zero order	(I)	$mol^{-1}L\,s^{-1}$
(B)	First order	(II)	$mol^{-2}L^2\,s^{-1}$
(C)	Second order	(III)	s^{-1}
(D)	Third order	(IV)	$mol\,L^{-1}\,s^{-1}$

Choose the **correct** answer from the options given below;

(1) A-IV, B-II, C-I, D-III
(2) A-IV, B-III, C-I, D-II
(3) A-IV, B-III, C-II, D-I
(4) A-I, B-II, C-III, D-IV

55. The correct IUPAC name of the following compound is :



(1) 3-ethyl-5-methylheptane
(2) 2,4-diethylhexane
(3) 3-methyl-5-ethylheptane
(4) 3,5-diethylhexane

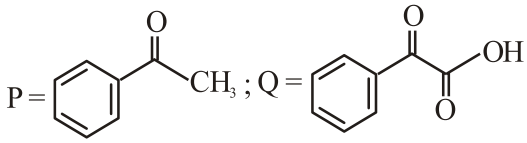
56. A bulb is rated at 150 watt, converting 8% energy into light. If energy of one photon is 4.42×10^{-19} J, how many photons are emitted by the bulb per second?

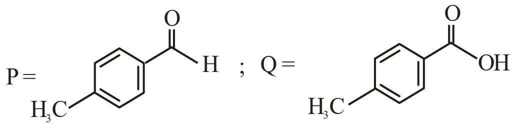
(1) 2.71×10^{19} (2) 4.06×10^{19}
(3) 27.2×10^{19} (4) 1.35×10^{19}

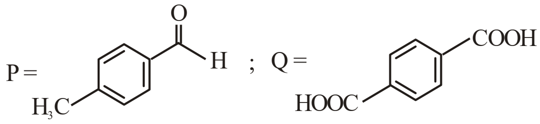
57. Methane reacts with steam at 1273 K in the presence of nickel catalyst to form :

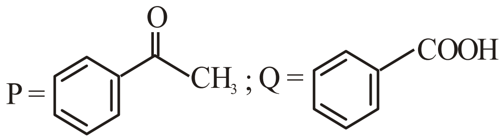
(1) CO and H_2O (2) CO and H_2
(3) CO_2 and H_2 (4) CO_2 and H_2O

58. Compound P (C_8H_8O) gives a red orange precipitate with 2,4-DNP reagent and it does not reduce Fehling's reagent. On drastic oxidation with chromic acid, P gives an aromatic product Q that produces effervescence on treating with aq. $NaHCO_3$. Compounds P and Q, respectively are:

(1) 

(2) 

(3) 

(4) 

59. Match List I with List II :

List-I		**List-I**	
A.	C_2H_4	I.	3 σ bonds, 2 π bonds
B.	C_2H_2	II.	3 σ bonds, one lone pair
C.	CH_4	III.	4 σ bonds
D.	NH_3	IV.	5 σ bonds, 1 π bond

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-II, D-I
(2) A-IV, B-I, C-III, D-II
(3) A-I, B-II, C-IV, D-III
(4) A-II, B-III, C-I, D-IV

60. The following two reactions give the same foul smelling product Z.

$$C_2H_5Cl \xrightarrow{X} Z$$

$$C_2H_5\overset{O}{\overset{\|}{C}}NH_2 \xrightarrow{Br_2,\,NaOH} Y \xrightarrow[\Delta]{CHCl_3/\text{ethanolic KOH}} Z$$

X and Z, respectively, are :

(1) X = AgCN; Z = C_2H_5NC
(2) X = KCN; Z = C_2H_5CN
(3) X = AgCN; Z = C_2H_5CN
(4) X = KCN; Z = C_2H_5NC

61. The number of hydrogen atoms present in 5.4 g of urea is :
(Given : Molar mass of urea : 60 g mol^{-1},
N_A : 6.022×10^{23} particles mol^{-1})

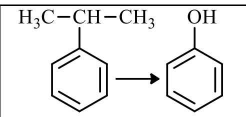
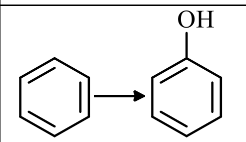
(1) 1.084×10^{23} (2) 1.084×10^{22}
(3) 2.168×10^{22} (4) 2.168×10^{23}

62. Identify the ***incorrect*** statement from the following :
(1) Nitrogen can form dπ-pπ bond with oxygen.
(2) $P(C_2H_5)_3$ and $As(C_6H_5)_3$ form dπ-dπ bond with transition metals.
(3) Phosphorus, arsenic and antimony show catenation property.
(4) Nitrogen can form pπ-pπ multiple bonds with itself.

63. Which one of the following is an ambidentate ligand?
(1) Ethane-1,2-diamine
(2) Ethylenediaminetetraacetate ion
(3) Thiocyanate
(4) Oxalate

64. The correct order of increasing metallic character of Na, Be, P, Mg and Si is :
(1) P < Si < Be < Mg < Na
(2) P < Si < Na < Mg < Be
(3) P < Mg < Be < Si < Na
(4) Be < Si < P < Mg < Na

65. Match **List-I** with **List-II**.

List-I		List-II	
(A)		(I)	(i) oleum (ii) NaOH (iii) H^+
(B)	$CH_3COOH \rightarrow CH_3CH_2OH$	(II)	(i) O_2 (ii) H_2O/H^+
(C)	$CH_3CH_2CH_2OH \rightarrow CH_3-\underset{\displaystyle OH}{\underset{\vert}{C}H}-CH_3$	(III)	(i) CH_3OH/H^+ (ii) H_2, catalyst
(D)		(IV)	(i) conc. H_2SO_4, Δ (ii) H^+/H_2O

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-II, B-III, C-IV, D-I
(3) A-II, B-IV, C-III, D-I
(4) A-I, B-III, C-IV, D-II

66. Although +3 oxidation state is most common in lanthanoids, cerium still shows +4 oxidation state because:
(1) After losing one more electron, it acquires $4f^{14}$ electronic configuration.
(2) Its nearest gas is Radon.
(3) Its atomic number is 61.
(4) After losing one more electron, it acquires $4f^0$ electronic configuration.

67. In the following reaction sequence, X and Z, respectively are:

$$CH_3CH_2CH_2{-}OH + PCl_5 \longrightarrow CH_3CH_2CH_2Cl + X + HCl$$
$$CH_3CH_2CH_2Cl \xrightarrow[\Delta]{\text{alc. KOH}} Y$$
$$Y \xrightarrow[(C_6H_5CO)_2O_2]{HBr} Z$$

(1) $X = POCl_3$; $Z = CH_3-\underset{\displaystyle Br}{\underset{\vert}{C}H}-CH_3$
(2) $X = POCl_3$; $Z = CH_3CH_2CH_2{-}Br$
(3) $X = H_3PO_3$; $Z = CH_3-\underset{\displaystyle Br}{\underset{\vert}{C}H}-CH_3$
(4) $X = H_3PO_3$; $Z = CH_3CH_2CH_2{-}Br$

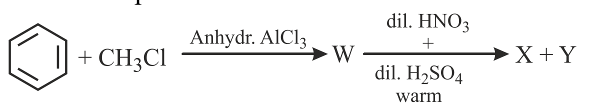
68. Match List I with List II:

List-I (Complex/ion)		List-I (Shape/geometry)	
A.	$[Pt(Cl_2)NH_3)_2]$	I.	Octahedral
B.	$[Co(NH_3)_6]Cl_3$	II.	Trigonal bipyramidal
C.	$[NiCl_4]^{2-}$	III.	Square planar
D.	$[Fe(CO)_5]$	IV.	Tetrahedral

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-III, B-I, C-IV, D-II
(3) A-IV, B-I, C-III, D-II
(4) A-I, B-III, C-IV, D-II

69. The functional group that can be identified through phthalein dye test is :
(1) Aldehyde
(2) Phenolic
(3) Carboxylic acid
(4) Alcohol

70. Two products X and Y are formed in the following reaction sequence.



The suitable method that can be used for the separation of products X and Y is:
(1) Fractional distillation
(2) Sublimation
(3) Differential extraction
(4) Continuous extraction

71. Identify the **correct** statements :

A. The molality of 2.5 g of ethanoic acid (Molar mass : $60\ g\ mol^{-1}$) in 75 g of benzene solution is 0.556 m.

B. The molarity of a solution containing 5 g of NaOH (molar mass : $40\ g\ mol^{-1}$) in 450 mL of solution is 0.278 M at 298 K.

C. Aquatic species are more comfortable in cold water.

D. The solubility of gas increases with decrease in pressure.

E. For a binary mixture of A and B, the number of moles of A and B are n_A and n_B respectively. The mole fraction of B will be $x_B = \frac{n_A}{n_A + n_B}$.

Choose the correct answer from the options given below :

(1) A, B and C only
(2) A and B only
(3) A and C only
(4) A, D and E only

72. During Lassaigne's test, the elements present in an organic compound are converted from :

(1) ionic form to ionic form
(2) covalent form to ionic form
(3) covalent form to covalent form
(4) ionic form to covalent form

73. A solution of copper sulphate is electrolysed for 10 minutes with a current of 1.5 amperes. The mass of copper deposited at cathode is :

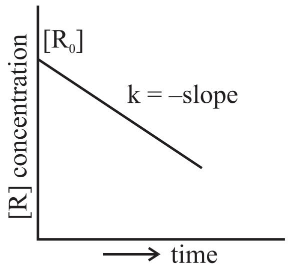
(Given : Molar mass of $Cu = 63\ g\ mol^{-1}$; $1\ F = 96487\ C\ mol^{-1}$)

(1) 1.7018 g
(2) 0.2938 g
(3) 2.4036 g
(4) 0.5876 g

74. At a certain temperature, T (K), during a process, 500 J is absorbed by the system and work of 200 J is done by the system. Then change in internal energy of the system is :

(1) 400 J (2) 300 J
(3) 700 J (4) 500 J

75. For a certain reaction R → Product, the plot of concentration [R] vs time has a negative slope as shown. The order of reaction is:



(1) 0
(2) 1
(3) 2
(4) 2.5

76. Identify the **correct** statement about ClF_3 from the following options:

(1) It has T-shaped geometry with two lone pairs on Cl atom.
(2) It has T-shaped geometry with three lone pairs on Cl atom.
(3) It has a trigonal pyramidal geometry with two lone pairs on Cl atom.
(4) It has a planar trigonal geometry with two lone pairs on Cl atom.

77. In a test tube containing a salt, a few drops of dilute H_2SO_4 was added, which gave colourless vapours having the smell of vinegar. The vapours turned the blue litmus paper red.
Identify the **correct** anion from the following:

(1) Sulphide, S^{2-}
(2) Sulphate, SO_4^{2-}
(3) Acetate, CH_3COO^-
(4) Carbonate, CO_3^{2-}

78. At 298 K, a certain buffer solution contains equal concentrations of X^- and HX, K_b for X^- is 10^{-10}. What is the pH of this buffer solution ?

(1) 2
(2) 4
(3) 6
(4) 10

79. Calculate emf of the half cell given below:

$Pt(s)\,|\,H_2(g, 2\ atm)\,|\,HCl(aq, 0.02\ M)$

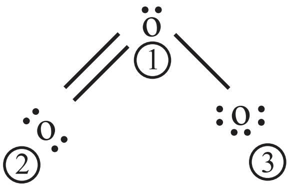
$E^\circ_{H_2/H^+} = 0\ V$

(Given : $\frac{2.303\ RT}{F} = 0.059, \log 2 = 0.3010$)

(1) –0.109 V (2) 0.035 V
(3) –0.035 V (4) 0.109 V

80. The calculated 'spin-only' magnetic moment of $Ti^{2+}(3d^2)$ is:
(1) 5.92 BM (2) 3.87 BM
(3) 2.84 BM (4) 4.90 BM

81. Identify the **incorrect** statement from the following:
(1) Carbon has the ability to form $p\pi - p\pi$ multiple bond with itself.
(2) ECl_3 (E = B and Al) is a monomer when E = B and a dimer when E = Al.
(3) The order of catenation property of Group 14 elements is C >> Si > Ge ≈ Sn.
(4) Oxygen exhibits only –2 oxidation state.

82.

The correct formal charges on oxygen atoms numbered 2, 1 and 3 respectively are :
(1) – 1, 0, + 1 (2) 0, + 1, – 1
(3) 0, 0, 0 (4) + 1, 0, – 1

83. Phenolphthalein is used as an indicator for the titration of sodium hydroxide solution against a standard solution of oxalic acid. The colour change that is observed at an alkaline pH close to the equivalence point during this titration is :
(1) pinkish red to yellow
(2) yellow to pinkish red
(3) pink to colourless
(4) colourless to pink

84. When 1 dm^3 of CO_2 gas is passed over hot coke, the volume of gaseous mixture after complete reaction at STP becomes 1.4 dm^3. The composition of the gaseous mixture at STP is:
(1) 0.8 dm^3 of CO, 0.8 dm^3 of CO_2
(2) 0.8 dm^3 of CO, 0.6 dm^3 of CO_2
(3) 0.6 dm^3 of CO, 0.8 dm^3 of CO_2
(4) 0.6 dm^3 of CO, 0.4 dm^3 of CO_2

85. The major product Z formed in the following sequence of reaction is:

$$C_2H_6 \xrightarrow[\text{UV light}]{Cl_2} \underset{\text{(monochlorinated product)}}{X} \xrightarrow{NH_3} Y \xrightarrow[\text{(ii) } H_2O]{\text{(i) } NaNO_2,\ HCl} Z$$

(1) $C_2H_5NO_2$
(2) $C_2H_5-N=N-OH$
(3) C_2H_5OH
(4) $C_2H_5NH_2$

86. Given below is an expression for the rate constant of a first order reaction occurring at a certain temperature, T (K).

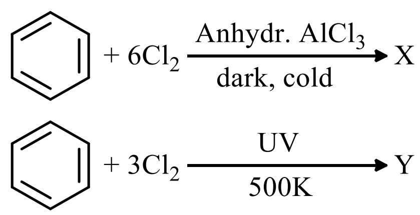
$$\ln k = 14.34 - \frac{1.25\times10^4}{T}$$

The energy of activation in kcal mol^{-1} for the reaction is:
(Given: k in s^{-1}, R = 1.987 cal mol^{-1} K^{-1})
(1) 24.84 (2) 14.34
(3) 18.63 (4) 12.42

87. Given below are certain reactions. Identify the reaction for which $K_p \neq K_c$.
(1) $H_2O(g) + CO(g) \rightleftharpoons H_2(g) + CO_2(g)$
(2) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
(3) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(4) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

88. Identify the **incorrect** statement from the following:
(1) The largest and the smallest species among Mg, Mg^{2+}, Al and Al^{3+} are Al and Mg^{2+}, respectively.
(2) The IUPAC name of the element with atomic number 107 is Unnilseptium.
(3) The similarity in behaviour of Li with Mg is referred to as 'diagonal relationship'.
(4) The oxidation state and covalency of Al in $[AlCl(H_2O)_5]^{2+}$ are 3 and 6 respectively.

89. Mixture of chloroform and acetone forms a solution with negative deviation from Raoult's law due to :
(1) increase in escaping tendency of molecules of each component.
(2) formation of hydrogen bonding between acetone and chloroform.
(3) stronger intermolecular forces between chloroform molecules than those between chloroform and acetone molecules.
(4) repulsive forces.

90. The number of chlorine atoms present in the organic products X and Y of the following reactions, respectively, are:


(1) 3 and 3 (2) 6 and 3
(3) 6 and 6 (4) 3 and 6

91. In angiosperms, root hairs arise from which one of the following regions of the root?

(1) The root cap zone

(2) The region of meristematic activity

(3) The region of elongation

(4) The region of maturation

92. In which one of the following, the ovules are ***not*** enclosed by an ovary wall and remain exposed?

(1) *Funaria*

(2) *Pinus*

(3) *Selaginella*

(4) *Wolffia*

93. In the *lac* operon, the *z* gene codes for:

(1) permease

(2) transacetylase

(3) beta-galactosidase

(4) the repressor of *lac* operon

94. Exploring molecular, genetic and species-level diversity for products of economic importance is called:

(1) Biofortification

(2) Bioremediation

(3) Bioprospecting

(4) Biomagnification

95. Match **List-I** with **List-II**:

List-I		List-II	
A.	Genetically modified organism	I.	*Agrobacterium tumefaciens*
B.	Thermostable DNA polymerase	II.	Bt cotton
C.	Ti plasmid	III.	*Thermus aquaticus*
D.	pBR322	IV.	*Escherichia coli*

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV

(2) A-II, B-I, C-IV, D-III

(3) A-I, B-IV, C-III, D-II

(4) A-I, B-II, C-IV, D-III

96. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Productivity	(I)	Gross primary productivity minus respiration losses
(B)	Net primary productivity	(II)	Rate of formation of new organic matter by consumers
(C)	Gross primary productivity	(III)	Rate of biomass production
(D)	Secondary productivity	(IV)	Rate of production of organic matter during photosynthesis

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-IV, D-II

(2) A-I, B-II, C-III, D-IV

(3) A-I, B-III, C-IV, D-II

(4) A-III, B-I, C-II, D-IV

97. Since the origin and diversification of life on Earth, there have been five episodes of mass extinction of species. How is the sixth, which is in progress, different from the previous episodes?

(1) The present net species extinction rate is zero.

(2) The current species extinction rates are nearly 10 times faster than that in previous episodes.

(3) The present species extinction rates are 100 to 1000 times faster than in the pre-human times.

(4) The current species extinction rates are far lower than those in previous episodes.

98. Alpha-helix is found in which level of protein structure?

(1) Secondary structure

(2) Tertiary structure

(3) Primary structure

(4) Quaternary structure

99. The main function of bulliform cells in grasses is :
(1) to make the leaf impermeable to fungal spores.
(2) to transport water.
(3) to perform photosynthesis.
(4) to minimize water loss during water stress.

100. Identify the correct sequence of steps in each cycle of Polymerase Chain Reaction:
(1) Extension → Annealing → Denaturation
(2) Annealing → Denaturation → Extension
(3) Denaturation → Extension → Annealing
(4) Denaturation → Annealing→ Extension

101. Match **List-I** with **List-II**:

List-I (Phase of cell cycle)		**List-II (Activity)**	
(A)	G_1 phase	(I)	Actual cell division occurs
(B)	S phase	(II)	Cell is metabolically active and continuously grows but does not replicate its DNA
(C)	G_2 phase	(III)	Synthesis of DNA occurs and the amount of DNA per cell doubles
(D)	M phase	(IV)	Proteins are synthesized while cell growth continues

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-II, D-III
(2) A-I, B-II, C-III, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-III, C-IV, D-I

102. Which of the following statements are **correct**?
A. The Amazon rainforest being cut and cleared for cultivation of soyabeans is an example of habitat loss.
B. Steller's sea cow and passenger pigeon became extinct due to over-exploitation by humans.
C. The Nile perch introduced into Lake Victoria in East Africa helped in population growth of cichlid fish in the lake.
D. Water hyacinth is an invasive species.
E. When a species becomes extinct, the plant and animal species associated with it are not affected.

Choose the **correct** answer from the options given below:
(1) A, B and E only
(2) A, B and D only
(3) C, D and E only
(4) B, C and D only

103. Which of the following statements are **correct** with reference to a transcription unit?
A. A transcription unit in DNA is defined primarily by three regions: promoter, structural gene and terminator.
B. The promoter is said to be located towards the 5'-end of the structural gene.
C. The promoter is a DNA sequence that provides binding site for RNA polymerase.
D. The promoter defines the template and coding strands.
E. The terminator is located towards the 3'-end of the coding strand and it defines the end of the process of transcription.

Choose the **correct** answer from the options given below:
(1) A, B, C and D only
(2) A, C, D and E only
(3) B, C, D and E only
(4) A, B, C, D and E

104. Which one of the following statements is ***not*** true about the universal rules of binomial nomenclature?
(1) Biological names are generally in Latin.
(2) Both the words in a biological name, when handwritten, are separately underlined and printed in italics.
(3) The specific epithet in the biological name starts with a small letter.
(4) The first word in the biological name represents the specific epithet, while the second component denotes the genus.

105. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Decomposition	(I)	Accumulation of dark coloured amorphous colloidal substance
(B)	Detritus	(II)	Release of inorganic nutrients by the activity of microbes in soil
(C)	Mineralisation	(III)	Breaking down of complex organic matter into inorganic substances
(D)	Humification	(IV)	Dead remains of plants and animals including fecal matter

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-I, D-II

(2) A-III, B-IV, C-II, D-I

(3) A-I, B-II, C-III, D-IV

(4) A-III, B-II, C-I, D-IV

106. Which one of the following is the site for active ribosomal RNA synthesis?

(1) Centrosome

(2) Chromatin

(3) Nucleolus

(4) Kinetochore

107. $2(C_{51}H_{98}O_6)+145O_2 \rightarrow 102CO_2+98H_2O+\text{energy}$

The Respiratory Quotient (RQ) of a biomolecule used for respiration, as per the above equation, would be:

(1) Between 0.5 and 0.95

(2) Less than 0.5

(3) 1.0

(4) Between 1.25 and 2

108. Match **List-I** with **List-II**:

List-I		List-II	
A.	Incomplete dominance	I.	Human skin colour
B.	Co-dominance	II.	Inheritance of flower colour in *Antirrhinum* sp.
C.	Pleiotropy	III.	Phenylketonuria disease in humans
D.	Polygenic inheritance	IV.	ABO blood groups

Choose the **correct** answer from the options given below:

(1) A-II, B-IV, C-III, D-I

(2) A-I, B-III, C-II, D-IV

(3) A-I, B-IV, C-III, D-II

(4) A-II, B-I, C-III, D-IV

109. Arrange the following steps of DNA fingerprinting in a correct sequence.

A. Isolation of DNA and its digestion by restriction endonucleases.

B. Hybridisation using a labelled VNTR probe.

C. Transferring of separated DNA fragments to synthetic membranes.

D. Detection of hybridised DNA fragments by autoradiography.

E. Separation of DNA fragments by electrophoresis.

Choose the **correct** answer from the options given below:

(1) A, B, D, C, E

(2) A, D, B, E, C

(3) A, E, C, B, D

(4) A, E, B, C, D

110. Which of the following statements are correct with reference to packaging of DNA helix?

A. Histones are organized to form a unit of eight molecules called histone octamer.

B. Histones are negatively charged basic proteins.

C. Histones are rich in the basic amino acid residues - lysine and arginine.

D. The positively charged DNA is wrapped around the histone octamer to form nucleosome.

E. The packaging of chromatin at higher levels requires an additional set of proteins called non-histone chromosomal proteins.

Choose the **correct** answer from the options given below:

(1) A, C and E only

(2) B, D and E only

(3) C, D and E only

(4) A, B and D only

111. Find the **incorrect** statement(s) about photosynthesis from the following:

A. The water splitting complex is associated with PS I.
B. C_4 plants use the C_3 pathway of CO_2 fixation as the main biosynthetic pathway.
C. In C_4 plants, photorespiration does not occur.
D. C_3 plants exhibit 'Kranz' anatomy.
E. ATP synthesis in chloroplast occurs through chemiosmosis.

Choose the answer from the options given below :

(1) B and C only
(2) B only
(3) B and E only
(4) A and D only

112. Arrange the following steps of somatic hybridisation in a correct sequence.

A. Digestion of cell walls.
B. Isolation of naked protoplasts.
C. Fusion of protoplasts to get hybrid protoplast.
D. Isolation of single cells from two different varieties of plants.
E. Growing of hybrid protoplast to form a new plant.

Choose the **correct** answer from the options given below:

(1) D, A, B, C, E
(2) E, B, A, D, C
(3) D, B, A, E, C
(4) E, A, B, C, D

113. Match **List I** with **List II**:

List-I		List-II	
A.	Conjunctive tissue	I.	Specialised cells in the vicinity of guard cells
B.	Casparian strips	II.	Endodermal cells rich in starch
C.	Subsidiary cells	III.	Tissue between xylem and phloem
D.	Starch sheath	IV.	Endodermal cells with suberin deposition

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-I, D-II
(2) A-III, B-IV, C-II, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-III, C-II, D-I

114. Which one of the following is ***not*** a characteristic of plant cells in the phase of elongation?

(1) New cell wall deposition
(2) Cell enlargement
(3) Increased vacuolation
(4) Large conspicuous nuclei

115. Match **List I** with **List II** :

List I (Growth Regulator)		List II (Function/Effect)	
A.	2,4-D	I.	Brewing industry
B.	GA_3	II.	Stimulation of stomatal closure
C.	Kinetin	III.	Herbicide
D.	ABA	IV.	Nutrient mobilisation

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-IV, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-IV, C-III, D-II
(4) A-I, B-II, C-IV, D-III

116. The enzyme required for carboxylation in the Calvin cycle is:

(1) Hexokinase
(2) PEP carboxylase
(3) RuBP carboxylase – oxygenase
(4) Carboxypeptidase

117. How many ATP and NADPH molecules are required to make one molecule of glucose through the Calvin pathway?

(1) 18 ATP and 12 NADPH
(2) 12 ATP and 18 NADPH
(3) 24 ATP and 18 NADPH
(4) 6 ATP and 12 NADPH

118. Which of the following floral formula is the correct floral formula of Solanaceae family ?

(1) $\oplus$ ⚥ $K_{(5)}\,C_{(5)}\,A_5\,\underline{G}_{(2)}$
(2) $\oplus$ ⚥ $K_5\,\overset{\frown}{C_{(5)}\,A_5}\,\underline{G}_{(2)}$
(3) $\oplus$ ⚥ $K_5\,C_5\,A_5\,\underline{G}_{(2)}$
(4) $\oplus$ ⚥ $K_{(5)}\,\overset{\frown}{C_{(5)}\,A_5}\,\underline{G}_{(2)}$

119 Which of the following is an *in situ* conservation method?

(1) Sacred Groves
(2) Wildlife Safari Parks
(3) Botanical Gardens
(4) Seed Banks

120. Which of the following statements are **not** true regarding restriction endonucleases?

A. They are called molecular scissors.

B. These are the enzymes responsible for restricting the growth of bacteriophages in *E. coli*.

C. They cut the DNA only at the centre of the palindromic sites.

D. They remove nucleotides only from the ends of DNA fragments.

E. They recognise specific palindromic base-pair sequences.

Choose the answer from the options given below:

(1) A and B only (2) A and E only

(3) D and E only (4) C and D only

121. In racemose inflorescence,

(1) the main axis terminates in a flower

(2) flowers are solitary

(3) the growth is limited

(4) flowers are borne in an acropetal succession

122. Arrange the following in the correct developmental sequence related to microsporogenesis :

A. Microspore tetrads

B. Sporogenous tissue

C. Pollen grains

D. Pollen mother cells

Choose the **correct** answer from the options given below:

(1) D, A, C, B

(2) B, D, A, C

(3) B, D, C, A

(4) A, D, C, B

123. Identify the correct statements about biomolecules.

A. Lipids are generally water soluble.

B. Proteins are polypeptides.

C. Polysaccharides are long chains of sugars.

D. Adenine and guanine are substituted pyrimidines.

E. Almost all enzymes are proteins.

Choose the **correct** answer from the options given below:

(1) B, D and E only

(2) B, C and E only

(3) A, B and C only

(4) C, D and E only

124. Which of the following statements are true with reference to the sex-determination in honeybees ?

A. An offspring formed from the union of a sperm and an egg, develops as a female (queen or worker).

B. An unfertilized egg develops as a male by parthenogenesis.

C. A male has half the number of chromosomes than that of a female.

D. Males produce sperms by meiosis.

E. Honeybees have a haplodiploid sex-determination system.

Choose the **correct** answer from the options given below:

(1) A, B, C and E only

(2) B, C, D and E only

(3) A, B, C and D only

(4) A, B, D and E only

125. Heterophyllous development in response to environment is an example of which of the following phenomena ?

(1) Redifferentiation

(2) Elasticity

(3) Dedifferentiation

(4) Plasticity

126. Which of the following statements are **correct** regarding amino acids ?

A. They are substituted methanes.

B. Serine is an aromatic amino acid.

C. Valine is a neutral amino acid.

D. Lysine is an acidic amino acid.

Choose the **correct** answer from the options given below:

(1) C and D only

(2) B and C only

(3) A and C only

(4) A and B only

127. "The Evil Quartet" of biodiversity loss includes which of the following?

(1) Over-exploitation; Alien species invasions; Air pollution; Co-extinctions

(2) Habitat loss and fragmentation; Air pollution; Water pollution; Co-extinctions

(3) Habitat loss and fragmentation; over-exploitation; Alien species invasions; Co-extinctions

(4) Over-exploitation; Alien species invasions; Soil pollution; Co-extinctions

128. Match **List I** with **List II** :

List-I (Process)		**List-II (Location)**	
A.	Glycolysis	I.	Inner mitochondrial membrane
B.	ETS	II.	Mitochondrial matrix
C.	Accumulation of protons	III.	Cytoplasm
D.	Krebs' cycle	IV.	Intermembrane space

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-I, D-III
(2) A-II, B-III, C-IV, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-I, B-IV, C-III, D-II

129. Which one of the following is a triploid cell?

(1) Synergid
(2) Central cell
(3) Zygote
(4) Primary endosperm cell

130. Which one of the following types of pollination brings genetically different types of pollen grains to the stigma ?

(1) Autogamy (2) Xenogamy
(3) Geitonogamy (4) Cleistogamy

131. Match **List-I** with **List-II**:

List-I (Placentation)		**List-II (Example)**	
(A)	Marginal	(I)	Mustard
(B)	Axile	(II)	Pea
(C)	Parietal	(III)	Marigold
(D)	Basal	(IV)	Lemon

Choose the **correct** answer from the options given below:

(1) A-II, B-IV, C-I, D-III
(2) A-I, B-III, C-II, D-IV
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-II, C-I, D-III

132. The main criteria used for Five Kingdom Classification proposed by R.H. Whittaker (1969) included:

A. Cell structure
B. Body organization
C. Presence of flagellum
D. Reproduction
E. Phylogenetic relationships

Choose the **correct** answer from the options given below:

(1) A, B, C, D and E
(2) B, C and D only
(3) A, B, D and E only
(4) A, B and E only

133. Match **List I** with **List II**:

List-I		**List-II**	
A.	Trypsin	I.	Intercellular ground substance
B.	Morphine	II.	Lectin
C.	Concanavalin A	III.	Enzyme
D.	Collagen	IV.	Alkaloid

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-II, D-I
(2) A-I, B-II, C-III, D-IV
(3) A-IV, B-III, C-II, D-I
(4) A-III, B-II, C-IV, D-I

134. Which of the following statements are correct with respect to DNA separation, isolation and visualization?

A. The cutting of DNA is done by molecular scissors.
B. The DNA fragments separate according to their size in an agarose gel, upon electrophoresis.
C. The separated DNA fragments can be seen without staining when exposed to UV light.
D. The separated DNA fragments, when stained with ethidium bromide, can be seen in visible light.

Choose the **correct** answer from the options given below :

(1) B and D only
(2) A and B only
(3) B and C only
(4) A and D only

135. Which one of the following disorders is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule ?

(1) Thalassemia
(2) Sickle-cell anaemia
(3) Phenylketonuria
(4) Haemophilia

136. Match List I with List II:

List I		List II	
A.	Cortisol	I.	Stimulates the formation of alveoli in mammary glands
B.	Aldosterone	II.	Produces anti-inflammatory reactions
C.	Cholecystokinin	III.	Stimulates reabsorption of Na^+ and water from renal tubule
D.	Progesterone	IV.	Stimulates secretion of pancreatic enzymes and bile juice

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-IV, D-I
(2) A-IV, B-II, C-I, D-III
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-III, C-I, D-IV

137. Arrange the following events occurring in Renin-Angiotensin mechanism in the correct order :

A. Increase in blood pressure and Glomerular filtration rate.
B. Reabsorption of Na^+ and water from distal parts of tubule due to Aldosterone.
C. Fall in Glomerular filtration rate.
D. Vasoconstriction by Angiotensin II and release of Aldosterone.
E. Renin converts Angiotensinogen into Angiotensin I, followed by Angiotensin II.

Choose the **correct** answer from the options given below :

(1) A, C, E, B, D
(2) C, A, B, D, E
(3) A, D, B, E, C
(4) C, E, D, B, A

138. In humans, respiration occurs in the following steps. Arrange these steps in the correct order.

A. Diffusion of O_2 and CO_2 between blood and tissues
B. Diffusion of O_2 and CO_2 across alveolar membrane
C. Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out
D. Cellular respiration
E. Transport of gases by the blood.

Choose the **correct** answer from the options given below:

(1) A, B, C, D, E (2) E, A, C, D, B
(3) C, B, E, A, D (4) C, A, B, E, D

139. The following are the stages of life cycle of *Plasmodium*. Arrange the stages in the proper order.

A. The parasites reproduce asexually in RBCs, bursting the cells.
B. The parasites reproduce asexually in liver cells, bursting the cells and releasing into blood.
C. Gametocytes develop in RBCs.
D. Sporozoites reach the liver through the blood.
E. Female mosquito injects sporozoites into humans during bite.

Choose the **correct** answer from the options given below:

(1) E, D, B, A, C (2) A, B, C, D, E
(3) C, A, B, D, E (4) E, C, D, B, A

140. Insertion of a foreign DNA at BamHI site in an *E. coli* cloning vector pBR322 results in the loss of antibiotic resistance towards:

(1) Ampicillin and tetracycline
(2) Ampicillin
(3) Tetracycline
(4) Gentamycin

141. The following reaction depicts the activity of a particular class of enzymes:

$$\underset{\text{(Substrate)}}{\overset{\;X\quad Y}{\overset{|\quad\;\, |}{C - C}}} \xrightarrow{E} \underset{\text{(Product)}}{X\text{–}Y} + \underset{\text{(Product)}}{C = C}$$

Identify the enzyme class 'E' from the following options:

(1) Transferases
(2) Isomerases
(3) Lyases
(4) Ligases

142. The specific receptors for neurotransmitters in a synapse are present on ______.

(1) Schwann cell
(2) Pre-synaptic membrane
(3) Myelin sheath
(4) Post-synaptic membrane

143. What is the probability of having children with ‘O’ blood group, where both mother and father are heterozygous for ‘A’ and ‘B’ blood group, respectively ?

(1) 25%
(2) 0%
(3) 75%
(4) 50%

144. Match **List I** with **List II**.

List I (Respiratory Volume)		List II (Capacity in mL)	
A.	ERV (Expiratory Reserve Volume)	I.	2500 – 3000 mL
B.	RV (Residual Volume)	II.	500 mL
C.	IRV (Inspiratory Reserve Volume)	III.	1000 – 1100 mL
D.	TV (Tidal Volume)	IV.	1100 – 1200 mL

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-I, D-II
(2) A-III, B-I, C-IV, D-II
(3) A-I, B-III, C-II, D-IV
(4) A-I, B-II, C-III, D-IV

145. Which of the following is ***not*** an example of convergent evolution?

(1) Flippers of penguins and dolphins
(2) Eyes of octopuses and mammals
(3) Fore limbs of whales and bats
(4) Wings of butterflies and birds

146. Male frogs can be distinguished from female frogs due to the presence of:

A. Bulging eyes
B. Vocal sacs
C. Webbed digits in feet
D. Copulatory pad on first digit of fore limbs
E. Olive green-coloured skin with dark irregular spots

Choose the **correct** answer from the options given below:

(1) B and C only (2) C and E only
(3) A and B only (4) B and D only

147. A group of researchers procured some fish-like animals and upon investigation the following characters were observed :

A. Endoskeleton was made of cartilage.
B. Ectoparasitic; as they were found attached on fish skin with their circular sucking mouth.
C. Paired fins and scales were absent, but 7 pairs of gill slits were present.

Which of the following species of animals did they consider to fit best with these characters?

(1) *Scoliodon* sp. (2) *Petromyzon* sp.
(3) *Exocoetus* sp. (4) *Branchiostoma* sp.

148. Match List I with List II with respect to chronology of evolution of life forms:

List I		List II	
A.	About 65 mya	I.	Jawless fish probably evolved
B.	About 500 mya	II.	The dinosaurs suddenly disappeared from the earth
C.	About 350 mya	III.	Seaweeds and few plants probably existed
D.	About 320 mya	IV.	Invertebrates were formed and became active

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-I, D-II
(2) A-I, B-II, C-III, D-IV
(3) A-II, B-IV, C-III, D-I
(4) A-II, B-IV, C-I, D-III

149. Match **List I** with **List II**.

List I		List II	
A.	Progestasert	I.	Barrier made of rubber used by females
B.	Multiload 375	II.	Oral contraceptive
C.	Diaphragm	III.	Hormone releasing IUD
D.	Saheli	IV.	Copper releasing IUD

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-II, C-I, D-III
(3) A-IV, B-III, C-I, D-II
(4) A-III, B-IV, C-II, D-I

150. The WBC count of a person's blood sample is 8000/cu.mm. How many eosinophils and lymphocytes would be in the same blood sample approximately?

(1) 300–500/cu.mm and 1200–1500/cu.mm, respectively
(2) 160–240/cu.mm and 1600–2000/cu.mm, respectively
(3) 300–500/cu.mm and 500–700/cu.mm, respectively
(4) 100–120/cu.mm and 160–200/cu.mm, respectively

151. Match **List-I** with **List-II**.

List-I (Drug)		List-II (Effect)	
(A)	Nicotine	(I)	Causes sense of euphoria and increased energy
(B)	Morphine	(II)	Stimulates adrenal gland to release catecholamines into blood circulation
(C)	Heroin	(III)	Effective sedative and painkiller
(D)	Cocaine	(IV)	A depressant; slows down body function

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-IV, D-I
(2) A-II, B-III, C-I, D-IV
(3) A-II, B-III, C-IV, D-I
(4) A-III, B-II, C-I, D-IV

152. The human protein named α-1 antitrypsin, obtained from transgenic animals, is used for the treatment of________.
(1) Emphysema
(2) Alzheimer's disease
(3) Rheumatoid arthritis
(4) Cystic fibrosis

153. Select the set of fishes which belong to the class Osteichthyes:
(1) Devil fish, Cuttlefish and Hagfish
(2) Saw fish, Fighting fish and Dog fish
(3) Starfish, Hagfish and Cuttlefish
(4) Flying fish, Angel fish and Fighting fish

154. Select the **incorrect** statements from the following:
A. Digestive system in Platyhelminthes is incomplete.
B. Bilateral symmetry is a characteristic feature of adult Echinoderms.
C. Pseudocoelom is possessed by Aschelminthes.
D. Notochord is persistent throughout life in the class Chondrichthyes.
E. Members of class Reptilia maintain a constant body temperature.
Choose the answer from the options given below:
(1) A and C only
(2) B and E only
(3) C and D only
(4) B and D only

155. Non-membrane bound cell organelles found in both prokaryotic and eukaryotic cells are __________.
(1) Ribosomes
(2) Lysosomes
(3) Centrosomes
(4) Mitochondria

156. Which of the following equations depicts Verhulst-Pearl logistic population growth?
(1) $dN/dt = rN (K + N / K)$
(2) $dN/dt = rN (K - N / K)$
(3) $dN/dt = rN (K - N / N)$
(4) $dN/dt = rN (K / K - N)$

157. Select the **incorrect** statements with reference to Rh grouping.
A. Erythroblastosis foetalis is a condition observed having foetus with Rh^{-ve} blood and mother with Rh^{+ve} blood.
B. Rh antigen is observed on RBCs in the majority of human beings.
C. Before blood transfusion, Rh group should also be matched.
D. Rh incompatibility is observed when a pregnant mother is Rh^{-ve} and the foetus is Rh^{+ve}.
E. Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the second child.
Choose the answer from the options given below:
(1) C and D only
(2) A and B only
(3) A and E only
(4) B and C only

158. Match **List-I** with **List-II**.

List-I (Bioactive molecules)		List-II (Importance)	
(A)	Streptokinase	(I)	Immunosuppressive agent
(B)	Statins	(II)	Removal of clots from the blood vessels
(C)	Lipases	(III)	Blood cholesterol-lowering agent
(D)	Cyclosporin A	(IV)	Detergent formulations

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-II, C-IV, D-I
(4) A-II, B-III, C-IV, D-I

159. Match List I with List II:

List I		List II	
A.	Molluscs	I.	Pulmonary respiration only
B.	Reptiles	II.	Branchial respiration
C.	Adult amphibians	III.	Cellular respiration
D.	*Amoeba*	IV.	Pulmonary and Cutaneous respiration

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-IV, D-III
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-II, C-I, D-IV

160. The sixth mutant codon of beta globin gene causing polymerization of Haemoglobin and change in RBC shape is ______.
(1) GUG (2) AUG
(3) GAG (4) CAG

161. Choose the correct statements regarding muscle contraction.
A. A motor neuron carries a signal sent by the Central Nervous System (CNS) to the sarcolemma of the muscle fibre.
B. The neural signal generates an action potential which causes the release of Ca^{++} into sarcoplasm.
C. Increase in Ca^{++} inactivates the actin for breaking cross bridges.
D. Actin binds to the myosin head to form a cross bridge.
E. Shortening of sarcomere takes place, by pulling actin filaments towards the centre of 'A' band.
Choose the **correct** answer from the options given below:
(1) C and D only
(2) A and B only
(3) C and E only
(4) A, B, D and E only

162. Which of the following statements are correct with reference to human endoskeleton ?
A. Human skull is monocondylic.
B. The joint between any two adjoining vertebrae is a cartilaginous joint.
C. In human beings, the number of cervical vertebrae is seven.
D. All ribs except the last 2 pairs are bicephalic.
E. The occipital bone of skull is articulated with atlas vertebra.
Choose the **correct** answer from the options given below:
(1) B and E only (2) B, C and E only
(3) C, D and E only (4) A, B and D only

163. Spermatogonia undergo a series of cell divisions to produce sperms. Select the correct statements from the following :
A. Spermatogonia always undergo meiotic cell division.
B. Primary spermatocytes divide mitotically to produce secondary spermatocytes.
C. Secondary spermatocytes, through their second meiotic division, produce haploid spermatids.
D. Spermatids produce spermatozoa through mitosis.
E. Spermatids transform into spermatozoa by spermiogenesis.
Choose the **correct** answer from the options given below :
(1) A and E only
(2) C and E only
(3) A, C and E only
(4) B, C and D only

164. The JGA (Juxta Glomerular Apparatus) is a special sensitive region formed by cellular modifications in ______ related to the same nephron.
(1) Distal convoluted tubule and efferent renal arteriole
(2) Proximal convoluted tubule and efferent renal arteriole
(3) Proximal convoluted tubule and afferent renal arteriole
(4) Distal convoluted tubule and afferent renal arteriole

165. Which one of the following is an appropriate example of 'sexual deceit'?
(1) Female wasp and fig
(2) *Ophrys* and bumblebee
(3) Sea anemone and clown fish
(4) Cuckoo and crow

166. Choose the correct statements regarding frog's anatomy:
A. Hepatic portal system is the special venous connection between liver and intestine.
B. There are twelve pairs of cranial nerves arising from the brain.
C. The ureters and oviducts open separately into the cloaca in female frogs.
D. Hind-brain consists of cerebellum, medulla oblongata and optic lobes.
E. Sinus venosus joins the right atrium of heart.
Choose the **correct** answer from the options given below:
(1) A, B and C only (2) B and D only
(3) B and C only (4) A, C and E only

167. Match **List I** with **List II** related to embryonic development at various months of pregnancy:

List I		List II	
A.	The foetus movement starts and hair appears on the head	I.	24 weeks of pregnancy
B.	The foetus develops limbs and digits	II.	20 weeks of pregnancy
C.	The foetus develops external genital organs	III.	8 weeks of pregnancy
D.	The foetus body is covered with fine hair; eyelids separate and eyelashes are formed	IV.	12 weeks of pregnancy

Choose the **correct** answer from the options given below :

(1) A-IV, B-II, C-III, D-I
(2) A-II, B-III, C-IV, D-I
(3) A-III, B-II, C-IV, D-I
(4) A-II, B-IV, C-III, D-I

168. In a population of grasshopper species, chromosome number of some members is 23 and some other members possess 24 chromosomes. The 23 and 24 chromosome-bearing members in this species are __________.

(1) females and males, respectively
(2) all males
(3) males and females, respectively
(4) all females

169. In which animal do haploid cells divide mitotically to produce gametes?

(1) Male honeybees
(2) Male earthworms
(3) Male frogs
(4) Male grasshoppers

170. Arrange the following cell layers/structure around the female gamete, from outer to inner side:

A. Zona pellucida
B. Perivitelline space
C. Corona radiata
D. Plasma membrane of ovum

Choose the **correct** answer from the options given below:

(1) D, B, A, C (2) C, A, D, B
(3) C, A, B, D (4) A, C, B, D

171. What is the reason behind production of large holes in 'Swiss Cheese' ?

(1) The production of large amount of CO_2 by *Propionibacterium sharmanii*
(2) The production of large amount of CO_2 by *Clostridium butylicum*
(3) The production of large amount of CO_2 and H_2 by lactic acid bacteria called *Lactobacillus*
(4) The production of large amount of CO_2 and H_2 by *Trichoderma polysporum*

172. The toxin proteins isolated from *Bacillus thuringiensis*, coded by which of the following genes would control cotton bollworms and corn borer, respectively?

(1) *cryIAc* and *cryIAb*
(2) *cryIAc* and *cryIIAb*
(3) *cryIIAb* and *cryIAc*
(4) *cryIAc* and *cryIIIAb*

173. Ecological pyramids represent the relationship between the organisms at different trophic levels and they are generally inverted for :

(1) Pyramid of biomass in grassland
(2) Pyramid of energy in pond ecosystem
(3) Pyramid of number in grassland
(4) Pyramid of biomass in sea

174. Choose the **correct** statement regarding GIFT to overcome infertility.

(1) Ova collected from a female donor are transferred to the uterus of an infertile female.
(2) Early embryos with up to 8 blastomeres are transferred to the uterus of an infertile female.
(3) Early embryos with up to 8 blastomeres are transferred into the fallopian tube of an infertile female.
(4) It is the transfer of an ovum collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilization and development.

175. Choose the correct statements regarding cell organelles and their inclusions.

A. The endomembrane system includes Golgi complex, endoplasmic reticulum and mitochondria.
B. Rough endoplasmic reticulum bears ribosomes on its surface.
C. Both mitochondria and plastids have circular DNA.
D. A network of microtubules, microfilaments and intermediate filaments present in the cytoplasm is called cytoskeleton.
E. Mitochondrion is a single membrane-bound structure.

Choose the **correct** answer from the options given below:
(1) A and B only
(2) A, B and C only
(3) C, D and E only
(4) B, C and D only

176. Select the correct statements regarding cell membrane in eukaryotic cell.
A. Membrane of human RBCs has approximately 52% protein.
B. Major phospholipids are arranged in a bilayer.
C. Extensions of the plasma membrane into the cell form mesosomes.
D. Tails towards the inner part of lipids are hydrophobic and thus protected from aqueous medium.
E. Glycocalyx is present on the outer surface of the plasma membrane.

Choose the **correct** answer from the options given below:
(1) C, D and E only (2) B, C and E only
(3) A, B and D only (4) A, C and E only

177. Match **List-I** with **List-II** related to muscular/skeletal system:

List-I		List-II	
(A)	Tetany	(I)	Inflammation of joints
(B)	Arthritis	(II)	Autoimmune disorder affecting neuromuscular junction
(C)	Myasthenia gravis	(III)	Wild contraction in muscle due to low Ca^{++} in body fluid
(D)	Muscular dystrophy	(IV)	Progressive degeneration of skeletal muscle

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-II, D-IV
(2) A-I, B-II, C-III, D-IV
(3) A-IV, B-III, C-II, D-I
(4) A-III, B-II, C-I, D-IV

178. Evolution of human appears parallel to the progressive development of brain and language skills. As such, the evolution of individual species in the sequence of their appearance is:
(1) *Ramapithecus* → *Homo habilis* → *Homo erectus* → *Neanderthal* → *Homo sapiens*
(2) *Homo sapiens* → *Ramapithecus* → *Homo habilis* → *Neanderthal* → *Homo erectus*
(3) *Homo habilis* → *Homo erectus* → *Ramapithecus* → *Neanderthal* → *Homo sapiens*
(4) *Neanderthal* → *Ramapithecus* → *Homo habilis* → *Homo erectus* → *Homo sapiens*

179. The flightless bird with forelimbs modified as paddle-like structures suited for swimming is known as :
(1) *Struthio*
(2) *Neophron*
(3) *Aptenodytes*
(4) *Psittacula*

180. Choose the correct statements regarding population interactions between two species.
A. In both parasitism and commensalism, only one species benefits and the other species is harmed.
B. Both species benefit in mutualism.
C. Both species benefit in commensalism.
D. In parasitism, only one species benefits and the other species is harmed.
E. In amensalism, one species is harmed and the other is unaffected.

Choose the **correct** answer from the options given below:
(1) B and E only
(2) A and B only
(3) B, D and E only
(4) A and D only

Test Booklet Code

ENGLISH **11**

NEET (UG)-2026

|| DATE: 03-05-2026 ||

ANSWER KEY

1.	(4)	46.	(2)	91.	(4)	136.	(3)
2.	(4)	47.	(4)	92.	(2)	137.	(4)
3.	(4)	48.	(3)	93.	(3)	138.	(3)
4.	(3)	49.	(3)	94.	(3)	139.	(1)
5.	(3)	50.	(4)	95.	(1)	140.	(3)
6.	(4)	51.	(2)	96.	(1)	141.	(3)
7.	(1)	52.	(4)	97.	(3)	142.	(4)
8.	(3)	53.	(4)	98.	(1)	143.	(1)
9.	(4)	54.	(2)	99.	(4)	144.	(1)
10.	(3)	55.	(1)	100.	(4)	145.	(3)
11.	(4)	56.	(1)	101.	(4)	146.	(4)
12.	(2)	57.	(3)	102.	(2)	147.	(2)
13.	(4)	58.	(4)	103.	(4)	148.	(4)
14.	(4)	59.	(2)	104.	(4)	149.	(1)
15.	(1)	60.	(1)	105.	(2)	150.	(2)
16.	(2)	61.	(4)	106.	(3)	151.	(3)
17.	(1)	62.	(4)	107.	(1)	152.	(1)
18.	(4)	63.	(3)	108.	(1)	153.	(4)
19.	(4)	64.	(1)	109.	(3)	154.	(2)
20.	(3)	65.	(2)	110.	(1)	155.	(1)
21.	(2)	66.	(4)	111.	(4)	156.	(2)
22.	(4)	67.	(2)	112.	(1)	157.	(3)
23.	(1)	68.	(2)	113.	(3)	158.	(4)
24.	(3)	69.	(2)	114.	(4)	159.	(1)
25.	(3)	70.	(1)	115.	(1)	160.	(1)
26.	(3)	71.	(1)	116.	(3)	161.	(4)
27.	(1)	72.	(2)	117.	(1)	162.	(2)
28.	(3)	73.	(2)	118.	(2)	163.	(2)
29.	(4)	74.	(2)	119.	(1)	164.	(4)
30.	(4)	75.	(1)	120.	(4)	165.	(2)
31.	(4)	76.	(1)	121.	(4)	166.	(4)
32.	(1)	77.	(3)	122.	(2)	167.	(2)
33.	(4)	78.	(2)	123.	(2)	168.	(3)
34.	(4)	79.	(4)	124.	(1)	169.	(1)
35.	(2)	80.	(3)	125.	(4)	170.	(3)
36.	(1)	81.	(4)	126.	(3)	171.	(1)
37.	(4)	82.	(2)	127.	(3)	172.	(1)
38.	(4)	83.	(4)	128.	(3)	173.	(4)
39.	(1)	84.	(2)	129.	(4)	174.	(4)
40.	(2)	85.	(3)	130.	(2)	175.	(4)
41.	(3)	86.	(1)	131.	(1)	176.	(3)
42.	(3)	87.	(2)	132.	(3)	177.	(1)
43.	(2)	88.	(1)	133.	(1)	178.	(1)
44.	(1)	89.	(2)	134.	(2)	179.	(3)
45.	(3)	90.	(3)	135.	(2)	180.	(3)

Hints & Solutions

Q1 Text Solution:

Time taken:

$6 \text{ min } 40 \text{ s} = 6 \times 60 + 40 = 400 \text{ s}$

Speed of light is taken as unity:

$c = 1$

$\text{Distance} = ct = 1 \times 400 = 400$

So, **option (3)**.

Q2 Text Solution:

A. Young's modulus:

$Y = \frac{FL}{A \Delta L}$

B. Compressibility:

$K = -\frac{1}{\Delta P}\left(\frac{\Delta V}{V}\right)$

C. Bulk modulus:

$B = -P\left(\frac{V}{\Delta V}\right)$

D. Poisson's ratio:

$\sigma = \frac{\Delta d/d}{\Delta L/L} = \frac{\Delta d}{\Delta L}\left(\frac{L}{d}\right)$

Q3 Text Solution:

Left rail is at higher potential $(+10 \text{ V})$.

So only the diodes allowing current left to right conduct.

Conducting branches: $(4\Omega) \text{ and} (2\Omega)$

Blocked branches: $(3\Omega) \text{ and} (5\Omega)$

$R_{eq} = \frac{4 \times 2}{4+2} = \frac{4}{3}\Omega$

$I = \frac{V}{R_{eq}} = \frac{10}{4/3} = \frac{15}{2} \text{ A}$

So, **option (2)**.

Q4 Text Solution:

Assuming uniform angular acceleration:

$$600 \text{ rpm} = 10 \text{ rps}, \quad 1200 \text{ rpm} = 20 \text{ rps}$$

Average frequency:

$$f_{avg} = \frac{10+20}{2} = 15 \text{ rev/s}$$

Number of revolutions in 10 s:

$$N = f_{avg} t = 15 \times 10 = 150$$

Q5 Text Solution:

For a simple pendulum in SHM:

$K.E. = \frac{1}{2}mv^2$

Velocity is zero at extreme positions, so K.E. is zero at:

$t = 0, \ \frac{T}{2}, \ T$

K.E. is always **positive** and maximum at mean position.

So, the correct graph is **option (3)**.

Q6 Text Solution:

Terminal voltage:

$V = E - Ir$

$V = 12 - (0.6)(2) = 12 - 1.2 = 10.8 \text{ V}$

Q7 Text Solution:

For a gas molecule:

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

At same temperature,

$$\frac{v_{\text{rms}}^{Ar}}{v_{\text{rms}}^{Cl}} = \sqrt{\frac{M_{Cl}}{M_{Ar}}}$$

$$= \sqrt{\frac{70}{40}} = \sqrt{\frac{7}{4}} = \frac{\sqrt{7}}{2}$$

The given mass ratio 2:1 is irrelevant here.

Q8 Text Solution:

For an equilateral prism:

$A = 60^\circ$

Since refracted ray (QR) is parallel to base, it makes 30° with the normal at each face:

$r_1 = r_2 = 30^\circ$

By reversibility/symmetry:

$e = i = 50^\circ$

Deviation:

$\delta = i + e - A = 50^\circ + 50^\circ - 60^\circ = 40^\circ$

Q9 Text Solution:

A. $E = h\nu \rightarrow$ **Energy of photon (IV)**

B. Diffraction and interference → **Wave nature of light (III)**

C. $\lambda = \frac{h}{p} \rightarrow$ **de Broglie wavelength (I)**

D. Compton effect → **Particle nature of light (II)**

NCERT states $E = h\nu$ for photon energy, Compton effect confirms particle-like behaviour, and $\lambda = h/p$ is the de Broglie wavelength.

Q10 Text Solution:

For hydrogen atom,

$$E=-\frac{ke^2}{2r}$$

So,

$$r=\frac{ke^2}{2|E|}$$

$$E=3.4\text{ eV}=3.4\times1.6\times10^{-19}\text{ J}$$

$$r=\frac{9\times10^9(1.6\times10^{-19})^2}{2\times3.4\times1.6\times10^{-19}}\approx2.1\times10^{-10}\text{ m}$$

Q11 Text Solution:

For the box to remain stationary relative to trolley, static friction must provide acceleration:

$$ma\le f_{s,\max}=\mu_s mg$$

$$a_{\max}=\mu_s g=0.12\times10=1.2\text{ m s}^{-2}$$

Mass cancels out.

Q12 Text Solution:

(C_1, C_2, C_3, C_4) are in series:

$C_s=\frac{10}{4}=2.5\ \mu F$

This is parallel with $C_5=2.5\ \mu F$:

$C_{eq}=2.5+2.5=5\ \mu F$

Each parallel branch gets (50 V).

$Q_{\text{series branch}}=2.5\times50=125\ \mu C$

In series, charge is same on each capacitor, so (C_1) to (C_4) each have $(125\ \mu C)$. Also,

$Q_5=2.5\times50=125\ \mu C$

Q13 Text Solution:

Work done:

$W=\Delta U=GMm\left(\frac{1}{R}-\frac{1}{2R}\right)$

$W=\frac{GMm}{2R}$

Since,

$g=\frac{GM}{R^2}\Rightarrow GM=gR^2$

$W=\frac{mgR}{2}$

Q14 Text Solution:

$\rho=\frac{m}{a^3}$

$a=9.0\text{ cm}=0.090\text{ m}$

$\rho=\frac{5.580}{(0.090)^3}=\frac{5.580}{0.000729}=7654\text{ kg m}^{-3}$

$\rho=7.654\times10^3\text{ kg m}^{-3}$

Here, least significant figures are in (9.0) cm = **2 significant figures**, so:

$X=7.7$

Q15 Text Solution:

For a ball thrown vertically upward, acceleration is constant downward (-g).

Taking upward as positive:

$v = u - gt$

So velocity starts positive, decreases linearly, becomes zero at the highest point, then becomes negative while falling.

Hence the correct (v)-(t) graph is **C only**.

Q16 Text Solution:

Total mechanical energy:

$$E=K+U=0.02\text{ J}$$

At equilibrium position, potential energy is minimum/taken zero, so:

$$K=0.02\text{ J}$$

$$\frac{1}{2}mv^2=0.02$$

Mass:

$$m=20\text{ g}=0.02\text{ kg}$$

$$v=\sqrt{\frac{2E}{m}}=\sqrt{\frac{2\times0.02}{0.02}}=\sqrt{2}\approx1.41\text{ m/s}$$

Q17 Text Solution:

In YDSE, for equal intensities:

$I=I_{\max}\cos^2\left(\frac{\phi}{2}\right)$

At path difference (λ), intensity is maximum, so:

$I_{\max}=K$

For path difference $(\lambda/3)$:

$\phi=\frac{2\pi}{\lambda}\cdot\frac{\lambda}{3}=\frac{2\pi}{3}$

$I=K\cos^2\left(\frac{\pi}{3}\right)=K\left(\frac{1}{2}\right)^2=\frac{K}{4}$

Q18 Text Solution:

The diode is **reverse biased during the positive half-cycle**, so the full positive half of input appears across the diode.

During the negative half-cycle, the diode is **forward biased**, so for an ideal diode:

$V_D = 0$

Hence (V_D) is a **positive half-wave only**

Q19 Text Solution:

For series LCR resonance:

$f = \frac{1}{2\pi\sqrt{LC}}$

$L = 1\ \mathrm{mH} = 10^{-3}\ \mathrm{H},\quad C = 0.1\ \mu F$

$= 10^{-7}\ \mathrm{F}$

$f = \frac{1}{2\pi\sqrt{10^{-10}}} = \frac{1}{2\pi\times10^{-5}} \approx 1.59\times10^{4}\ \mathrm{Hz}$

$= 15.9\ \mathrm{kHz}$

Q20 Text Solution:

A is true: In interference/diffraction, energy is only redistributed. Dark fringes have less energy and bright fringes have more energy, so conservation of energy is obeyed.

B is false: Interference and diffraction are wave phenomena, not exclusive to light. Sound waves, water waves, etc. also show them.

Q21 Text Solution:

Given:

$y = 2\cos 2\pi(10t - 0.0080x + 0.35)$

Phase difference for distance Δx:

$\Delta\phi = 2\pi(0.0080)\Delta x$

$\Delta x = 0.5\ \mathrm{m} = 50\ \mathrm{cm}$

$\Delta\phi = 2\pi(0.0080)(50) = 0.8\pi\ \mathrm{rad}$

Q22 Text Solution:

Resultant force:

$F = \sqrt{8^2 + 6^2} = 10\ \mathrm{N}$

$a = \frac{F}{m} = \frac{10}{5} = 2\ \mathrm{m\ s^{-2}}$

Direction with 8 N force:

$\tan\theta = \frac{6}{8} = \frac{3}{4}$

Q23 Text Solution:

$U_i = \frac{1}{2}CV^2$

Here $C = 200\ pF,\ V = 100\ V$

When a charged capacitor is connected to an uncharged capacitor, charge is redistributed until both reach a common potential.

Since the capacitors are identical, the charge will split **exactly in half** between them.

So final voltage across each:

$V_f = \frac{V}{2}$

$U_f = \frac{1}{2}\left(C\right)\left(\frac{V}{2}\right)^2 + \frac{1}{2}\left(C\right)\left(\frac{V}{2}\right)^2 = \frac{1}{4}CV^2$

Energy lost:

$\Delta U = \frac{1}{2}CV^2 - \frac{1}{4}CV^2 = \frac{1}{4}CV^2$

$= \frac{1}{4}\left(200\times10^{-12}\right)\left(100\right)^2 = 0.5\times10^{-6}\ \mathrm{J}$

Q24 Text Solution:

$P = \frac{W}{t} = \frac{mgh}{t}$

$P = \frac{1000\times9.8\times20}{10} = 19600\ \mathrm{W}$

$19600\ \mathrm{W} = 19.6\ \mathrm{kW}$

Q25 Text Solution:

Given: $20\ \mathrm{VSD} = 16\ \mathrm{MSD}$

$1\ \mathrm{VSD} = \frac{16}{20}\ \mathrm{MSD} = 0.8\ \mathrm{mm}$

$\text{Least count} = 1\ \mathrm{MSD} - 1\ \mathrm{VSD} = 1 - 0.8$

$= 0.2\ \mathrm{mm}$

$0.2\ \mathrm{mm} = 0.02\ \mathrm{cm}$

Q26 Text Solution:

Distance fallen:

$s = \frac{1}{2}gt^2$

So greater reaction time means greater distance.

Times:

$B = 0.22\ >\ E = 0.21\ >\ A = 0.20\ >$
$D = 0.19\ >\ C = 0.18$

Q27 Text Solution:

Total wire resistance $= 4\Omega$, so each side of square:

$R_{side} = 1\Omega$

Battery is across opposite corners (A) and (C). By symmetry, (B) and (D) are at the same potential, so no current flows through the 2Ω diagonal resistor.

Two equal paths:

$A - B - C = 1 + 1 = 2\Omega$

$A - D - C = 1 + 1 = 2\Omega$

These are in parallel:

$R_{eq} = \frac{2\times2}{2+2} = 1\Omega$

$I = \frac{V}{R} = \frac{2}{1} = 2\ \mathrm{A}$

Q28 Text Solution:

For a heater, resistance remains constant:

$R = \frac{V^2}{P} = \frac{220^2}{400} = 121\Omega$

At $(200\ \mathrm{V})$:

$P' = \frac{V'^2}{R} = \frac{200^2}{121} \approx 331\ \mathrm{W}$

Q29 Text Solution:

Using NCERT formulae: $B=\frac{\mu_0 NI}{2R}$ and magnetic moment (m=NIA).

$$\left[I=\frac{2RB}{\mu_0 N}=\frac{2(0.05)(3.14\times10^{-3})}{(4\pi\times10^{-7})(100)}=2.5\ A\right]$$

$$\left[m=NI\pi R^2=100(2.5)\pi(0.05)^2\right.$$

$$\left.\approx 2\ A\ m^2\right]$$

Q30 Text Solution:

For rectangular loop moving out of magnetic field:

$\varepsilon=Blv$

Here, velocity is normal to the **shorter side**, so

$l=3\ cm=0.03\ m$

$\varepsilon=0.3\times0.03\times0.02=1.8\times10^{-4}\ V$

Q31 Text Solution:

Nuclear radius:

$$R=R_0A^{1/3}$$

So nuclear volume:

$$V\propto R^3\propto A$$

Hence **A is false**, **B is true**.

Mass defect is the difference between the mass of a nucleus and the total mass of its constituent protons and neutrons. Hence **C is false**, **D is true**.

Q32 Text Solution:

$R=R_0A^{1/3}$,

so nuclear volume $=\frac{4}{3}\pi R_0^3A$.

$\rho=\frac{m}{V}=\frac{m}{\frac{4}{3}\pi R_0^3A}$

$A=\frac{m}{\rho\cdot\frac{4}{3}\pi R_0^3}$

$$A=\frac{19.926\times10^{-27}}{(2.29\times10^{17})\left(\frac{12.56}{3}\right)(1.2\times10^{-15})^3}\approx12$$

Q33 Text Solution:

Time period:

$$T=\frac{60}{30}=2s$$

For simple pendulum:

$$T=2\pi\sqrt{\frac{L}{g}}$$

$$L=\frac{gT^2}{4\pi^2}=\frac{9.8\times4}{4\times9.8}=1\ m$$

So, length = $1\ m$.

Q34 Text Solution:

Using first law of thermodynamics:

$\frac{dQ}{dt}=\frac{dU}{dt}+\frac{dW}{dt}$

$100=\frac{dU}{dt}+75$

$\frac{dU}{dt}=25\ W$

Q35 Text Solution:

Here (m) is linear mass density, so total mass:

$M=mL$

$L=2\pi R\Rightarrow R=\frac{L}{2\pi}$

Moment of inertia of ring about diameter:

$I_C=\frac{MR^2}{2}$

Axis (yy') is tangent and parallel to diameter, so by parallel axis theorem:

$I=I_C+MR^2=\frac{3}{2}MR^2$

$I=\frac{3}{2}(mL)\left(\frac{L}{2\pi}\right)^2=\frac{3mL^3}{8\pi^2}$

Q36 Text Solution:

For shunt:

$S=\frac{I_gG}{I-I_g}$

$I_g=1\ mA=10^{-3}\ A,\quad G=100\Omega,\quad I=10\ A$

$S=\frac{10^{-3}\times100}{10-10^{-3}}=\frac{0.1}{9.999}\approx0.01\Omega$

Q37 Text Solution:

A metre bridge is a Wheatstone bridge. Interchanging the cell and galvanometer does not change the balance condition.

So, on two sides of the balance point, galvanometer deflects in opposite directions, and at balance:

$G=0$ (Deflection in galvanometer is zero)

Q38 Text Solution:

For AC starting from zero:

$$i=i_0\sin\omega t$$

Peak value occurs when:

$$\sin\omega t=1\Rightarrow\omega t=\frac{\pi}{2}$$

$$\frac{2\pi}{T}t=\frac{\pi}{2}$$

So time taken is one-fourth of time period:

$t = \frac{T}{4} = \frac{1}{4f}$

$t = \frac{1}{4 \times 60} = \frac{1}{240}\ \mathrm{s}$

Q39 Text Solution:

For a uniformly distributed current in a solid wire:

Inside $(\mathrm{r} < \mathrm{a})$:

$\mathrm{B} = \frac{\mu_0\ \mathrm{Ir}}{2\pi \mathrm{a}^2}$

So, $(\mathrm{B} \propto \mathrm{r})$ linear increase.

Outside $(\mathrm{r} > \mathrm{a})$:

$\mathrm{B} = \frac{\mu_0 \mathrm{I}}{2\pi \mathrm{r}}$

Q40 Text Solution:

A is true: In forward bias, after threshold/cut-in voltage, diode current increases significantly.

B is false: This current is **forward current**, not reverse saturation current. Reverse saturation current is the small current in reverse bias.

Q41 Text Solution:

- **A true:** In electrostatic equilibrium, electric field inside a conductor is zero.
- **B false:** Electric field just outside surface depends on surface charge density:

$$E = \frac{\sigma}{\varepsilon_0}$$

- **C true:** Excess charge resides only on the surface of a charged conductor.
- **D true:** Electric field at conductor surface is normal to the surface.
- **E false:** Potential inside a charged conductor is **constant**, not necessarily zero.

Q42 Text Solution:

Photoelectric effect occurs only if:

$\frac{hc}{\lambda} \geq \phi$

Threshold wavelength:

$\lambda_0 = \frac{hc}{\phi}$

$\phi = 6.6\ \mathrm{eV} = 6.6 \times 1.6 \times 10^{-19}\ \mathrm{J}$

$\lambda_0 = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{6.6 \times 1.6 \times 10^{-19}} = 1.875 \times 10^{-7}\ \mathrm{m}$

$= 187.5\ \mathrm{nm}$

Wavelength greater than (187.5 nm) will **not** produce photoelectric effect.

$200\ \mathrm{nm} > 187.5\ \mathrm{nm}$

Q43 Text Solution:

For a **concave lens**, a ray parallel to the principal axis diverges after refraction and appears to come from the **same-side focus**.

Q44 Text Solution:

At depth *h*,

$P_{\text{absolute}} = P_{\text{atm}} + \rho g h$

Given:

$100\ \mathrm{atm} = 1\ \mathrm{atm} + \rho g h$

$\rho g h = 99\ \mathrm{atm} = 99 \times 10^5\ \mathrm{Pa}$

$h = \frac{99 \times 10^5}{1000 \times 10} = 990\ \mathrm{m}$

So, **maximum depth =** $990\ \mathrm{m}$.

Q45 Text Solution:

- **Microwave** → Klystron valve or magnetron valve
- **Visible light** → Electrons jump from higher to lower energy levels
- **Gamma rays** → Radioactive decay of nucleus
- **Infra-red rays** → Vibrations of atoms and molecules

Q46 Text Solution:

Nitriles $(\mathrm{R} - \mathrm{C} \equiv \mathrm{N})$ reduce to primary amines $(\mathrm{R} - \mathrm{CH_2\ NH_2})$ with:

A. $(\mathrm{LiAlH_4}\ / \mathrm{H_2O})$—yes

C. $(\mathrm{H_2} / \mathrm{Ni})$—yes

D. $(\mathrm{Na(Hg)} / \mathrm{C_2H_5\ OH})$—yes

B. $(\mathrm{Sn} + \mathrm{HCl})$ is not the standard reduction of nitrile to primary amine; it is associated with other reductions/Stephen-type reduction.

E. $(\mathrm{Br_2} / \mathrm{aq.\ \ NaOH})$ is Hofmann bromamide reaction for amides, not nitriles.

Q47 Text Solution:

A. V_2O_5 → Preparation of H_2SO_4 from SO_2
B. Fe → Preparation of ammonia from N_2/H_2 mixture
C.$PdCl_2$ → Oxidation to ethanal
D. Ni complex → Polymerisation of alkynes

Q48 Text Solution:

$\Delta H^\circ = \Delta U^\circ + \Delta n_g RT$

$\Delta n_g = 2 - \left(2+1\right) = -1$

$\Delta H^\circ = -10 - \frac{(8.31)(298)}{1000} = -12$

$.476\ \text{kJ mol}^{-1}$
$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$
$= -12.476 -$

$298 \times \left(-44\right)/1000$

$= +0.6356\ \text{kJ mol}^{-1}$
Since $\Delta G^\circ > 0$, reaction is **non-spontaneous**.

Q49 Text Solution:

Use *l*-values:
$l = 0 \rightarrow s,\ \ l = 1 \rightarrow p,\ \ l = 2 \rightarrow d,\ \ l = 3$
$\rightarrow f$
A: *n*=2, *l*=1 → **2p** → II
B: *n*=4, *l*=0 → **4s** → III
C: *n*=5, *l*=3 → **5f** → IV
D: *n*=3, *l*=2 → **3d** → I
So, **A-II, B-III, C-IV, D-I**.

Q50 Text Solution:

For
$BiO(OH)(s) \rightleftharpoons BiO^+ + OH^-$

$K = [BiO^+][OH^-]$
Let solubility = *s*, so:

$s^2 = 4 \times 10^{-10}$

$s = [OH^-] = 2 \times 10^{-5}$

$pOH = -\log(2 \times 10^{-5}) = 5 - 0.301 = 4$
$.699$

$pH = 14 - 4.699 = 9.301$
So, **pH = 9.301**.

Q51 Text Solution:

DNA possesses a **double-stranded helical structure** and contains **thymine** as one of its four nitrogenous bases.
RNA is generally **single-stranded** and contains **uracil instead of thymine**.

Q52 Text Solution:

$CH_3OCH_2CH_2CH_3$
and
$CH_3CH_2OCH_2CH_3$
Both are ethers with the same molecular formula $C_4H_{10}O$, but different alkyl groups on the two sides of oxygen.
So they are **metamers**.

Q53 Text Solution:

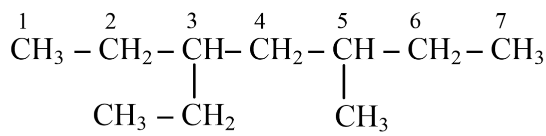
A. $[Pt(NH_3)_2Cl_2]$ → **Geometrical isomerism**
B. $[Co(en)_3]^{3+}$ → **Optical isomerism**
C. $[Co(NH_3)_5NO_2]Cl_2$ → **Linkage isomerism**
D. $[Cr(H_2O)_6]Cl_3$ → **Solvate/Hydrate isomerism**

Q54 Text Solution:

Unit of rate constant:
$k = (\text{mol L}^{-1})^{1-n}\ \text{s}^{-1}$
So:
A. Zero order → $\text{mol L}^{-1}\ \text{s}^{-1} \rightarrow \text{IV}$
B. First order → $\text{s}^{-1} \rightarrow \text{III}$
C. Second order → $\text{L mol}^{-1}\ \text{s}^{-1} \rightarrow \text{I}$
D. Third order → $\text{L}^2\ \text{mol}^{-2}\ \text{s}^{-1} \rightarrow \text{II}$

Q55 Text Solution:

Longest chain has **7 carbons**, so parent hydrocarbon is **heptane**.
Substituents:
3-ethyl, 5-methyl
Since locants are (3,5) from both directions, numbering is chosen so the substituent written first alphabetically (**ethyl**) gets the lower number.



So correct name is:
3-ethyl − 5-methylheptane

Q56 Text Solution:

Power converted into light:
$150 \times \frac{8}{100} = 12 \text{ J s}^{-1}$
Energy of one photon:
4.42×10^{-19} J
Number of photons per second:
$\frac{12}{4.42 \times 10^{-19}} = 2.71 \times 10^{19}$

Q57 Text Solution:

Methane reacts with steam over Ni catalyst at high temperature:
$$CH_4 + H_2O \xrightarrow[1273 \text{ K}]{\text{Ni}} CO + 3H_2$$
This is **steam reforming of methane**.

Q58 Text Solution:

P is **acetophenone**:
$C_6H_5 COCH_3$
It gives 2,4-DNP test because it has a carbonyl group, and it does not reduce Fehling's reagent because it is a ketone.
On drastic oxidation with chromic acid:
$C_6H_5 COCH_3 \rightarrow C_6H_5 COOH$
Q is **benzoic acid**, which gives effervescence with $NaHCO_3$ due to CO_2 evolution.

Q59 Text Solution:

A. C_2H_4 : $5\sigma, 1\pi$
B. C_2H_2 : $3\sigma, 2\pi$
C. CH_4 : 4σ
D. NH_3 : 3σ, one lone pair

Q60 Text Solution:

$X = \text{AgCN}, \quad Z = C_2H_5 NC$
Reason:
$C_2H_5Cl + AgCN \rightarrow C_2H_5 NC$
AgCN gives **isocyanide**, which has foul smell.
Also,
$$C_2H_5 CONH_2 \xrightarrow{Br_2 / NaOH} C_2H_5 NH_2$$
Then ethylamine gives carbylamine reaction:
$C_2H_5 NH_2 + CHCl_3 + \text{alc. KOH}$
$\rightarrow C_2H_5 NC$
So (Z) is **ethyl isocyanide**.

Q61 Text Solution:

Urea: $NH_2 CONH_2$
Each urea molecule has **4 H atoms**.
Moles of urea:
$\frac{5.4}{60} = 0.09$ mol
Number of H atoms:
$0.09 \times 6.022 \times 10^{23} \times 4 = 2.168 \times 10^{23}$

Q62 Text Solution:

Nitrogen is a **2nd-period element** and has **no vacant d-orbitals**, so it cannot form a $d\pi - p\pi$ bond with oxygen.
Nitrogen forms $p\pi - p\pi$ multiple bonding, e.g. in:
$N \equiv N$

Q63 Text Solution:

SCN^- is an **ambidentate ligand** because it can coordinate through either:
S
or
N
So it forms linkage isomers: thiocyanato-S and isothiocyanato-N.

Q64 Text Solution:

$P < Si < Be < Mg < Na$
Metallic character **decreases left to right** across a period and **increases down a group**.
So among these, **P is least metallic** and **Na is most metallic**.

Q65 Text Solution:

A-II: Cumene → phenol by oxidation with O_2 followed by acid hydrolysis.
B-III: CH_3COOH → CH_3CH_2OH via esterification with CH_3OH/H^+, then hydrogenation.
C-IV: 1-propanol → 2-propanol by dehydration to propene, then acid-catalysed hydration.
D-I: Benzene → phenol by sulphonation with oleum, NaOH fusion, then acidification.
So, **A-II, B-III, C-IV, D-I**.

Q66 Text Solution:

Cerium: $Ce, \quad Z = 58$
$Ce^{3+} = [Xe], 4f^1$
$On\ losing\ one\ more\ electron:$
$Ce^{4+} = [Xe], 4f^0$
So, Ce^{4+} is stable because it attains an empty $4f^0$ configuration.

Q67 Text Solution:

$$CH_3\,CH_2\,CH_2\,OH + PCl_5$$
$$\rightarrow CH_3\,CH_2\,CH_2\,Cl + POCl_3 + HCl$$

So, **X = $POCl_3$**.

Then:

$$CH_3\,CH_2\,CH_2\,Cl \xrightarrow{\text{alc. KOH}, \Delta} CH_3\,CH$$
$$= CH_2$$

Propene reacts with HBr in presence of benzoyl peroxide by **anti-Markovnikov addition**:

$$CH_3\,CH = CH_2 + HBr$$
$$\rightarrow CH_3\,CH_2\,CH_2\,Br$$

So, **Z = $CH_3CH_2CH_2Br$**.

Answer: X = $POCl_3$, Z = $CH_3CH_2CH_2Br$.

Q68 Text Solution:

A. $[PtCl_2(NH_3)_2]$ → **Square planar**

B. $[Co(NH_3)_6]\,Cl_3$ → **Octahedral**

C. $[NiCl_4]^{2-}$ → **Tetrahedral**

D. $[Fe(CO)_5]$ → **Trigonal bipyramidal**

Q69 Text Solution:

Phthalein dye test is used to identify **phenolic –OH** group.

Phenol reacts with phthalic anhydride in presence of conc. $H_2\,SO_4$ to form phenolphthalein-type dye, which gives pink colour in alkaline medium.

Q70 Text Solution:

First step:

$$C_6H_6 \xrightarrow[AlCl_3]{CH_3\,Cl} C_6H_5\,CH_3$$

This forms **toluene**.

On nitration, toluene gives mainly:

o-nitrotoluene and p-nitrotoluene

These can be separated by **fractional distillation** due to difference in their boiling points.

Q71 Text Solution:

A. Correct:

$$\text{m} = \frac{2.5/60}{75/1000} = 0.556 \text{ m}$$

B. Correct:

$$\text{M} = \frac{5/40}{0.450} = 0.278 \text{ M}$$

C. Correct: Cold water dissolves more O_2, so aquatic species are more comfortable.

D. Incorrect: Gas solubility **increases with increase in pressure**.

E. Incorrect:

$x_B = \frac{n_B}{n_A+n_B} not \frac{n_A}{n_A+n_B}$.

Q72 Text Solution:

In Lassaigne's test, covalently bonded elements in organic compounds like N, S, X are fused with sodium and converted into ionic sodium salts, e.g.

$$Na + C + N \rightarrow NaCN$$

So detection becomes easier in aqueous extract.

Q73 Text Solution:

For copper deposition:

$$Cu^{2+} + 2e^- \rightarrow Cu$$
$$Q = It = 1.5 \times 10 \times 60 = 900 \text{ C}$$
$$m = \frac{MQ}{nF} = \frac{63 \times 900}{2 \times 96487} \approx 0.2938 \text{ g}$$

Q74 Text Solution:

Heat absorbed by system:

$q = +500$ J

Work done by system:

$w = -200$ J

So,

$$\Delta U = q + w = 500 - 200 = 300 \text{ J}$$

Q75 Text Solution:

For a **zero-order reaction**:

$$[R] = [R_0] - kt$$

So the plot of ([R]) vs time is a **straight line with negative slope**.

$$\text{slope} = -k$$

Hence, the reaction is **zero order**.

Q76 Text Solution:

ClF_3

Central Cl has:

- 3 bond pairs
- 2 lone pairs

So VSEPR type is:

$AB_3\,E_2$

Electron-pair geometry is trigonal bipyramidal, but molecular shape is **T-shaped**.

Q77 Text Solution:

Dilute H_2SO_4 reacts with acetate salt to liberate **acetic acid vapours**, which have a **vinegar smell** and turn blue litmus red.

$$2\,CH_3COONa + H_2SO_4 \rightarrow Na_2SO_4 + 2\,CH_3COOH$$

This is the characteristic preliminary test for acetate ion.

Q78 Text Solution:

Given:

$K_b\left(X^-\right) = 10^{-10}$

So,

$pK_b = 10$

For conjugate acid-base pair:

$pK_a + pK_b = 14$

$pK_a = 14 - 10 = 4$

Since $[X^-] = [HX]$,

$pH = pK_a = 4$

Q79 Text Solution:

$H_2 \rightarrow 2H^+ + 2e^-$

$E_{cell} = E^\circ{}_{cell} - \frac{0.059}{2}\log\frac{(2\times10^{-2})^2}{2}$

$= 0 - \frac{0.059}{2}\log\frac{4\times10^{-4}}{2}$

$= \frac{-0.059}{2}[\log 2 - 4]$

$= -0.0295\left[0.3010 - 4\right]$

$= +0.109$

Q80 Text Solution:

For Ti^{2+}:

$3d^2$

Number of unpaired electrons:

n = 2

Spin-only magnetic moment:

$\mu = \sqrt{n(n+2)}$

$\mu = \sqrt{2(2+2)} = \sqrt{8} = 2.84\ \text{BM}$

Q81 Text Solution:

Oxygen can also show:

-1 in peroxides, $-\frac{1}{2}$ in superoxides, and positive oxidation states in compounds with fluorine, e.g. OF_2.

Other statements are correct.

Q82 Text Solution:

For the given ozone structure:

- O atom **2** is double bonded → formal charge (0)
- Central O atom **1** has one double bond and one single bond → formal charge +1
- O atom **3** is single bonded with three lone pairs → formal charge –1

So, for atoms **2, 1, 3** respectively:

0, +1, –1

Q83 Text Solution:

Phenolphthalein is **colourless in acidic solution** and **pink in alkaline solution**.

In oxalic acid vs NaOH titration, near equivalence point the medium becomes slightly alkaline, so the endpoint is:

$\text{colourless} \rightarrow \text{pink}$

Q84 Text Solution:

Reaction with hot coke:

$CO_2 + C \rightarrow 2\,CO$

Let x dm^3 of CO_2 react.

Remaining $CO_2 = 1 - x$

CO formed = $2x$

$$(1-x) + 2x = 1.4$$

$$1 + x = 1.4 \Rightarrow x = 0.4$$

So,

$$CO = 0.8\ dm^3,\quad CO_2 = 0.6\ dm^3$$

Q85 Text Solution:

Sequence:

C_2H_6 → **C_2H_5Cl** by chlorination

$C_2H_5Cl + NH_3$ → **$C_2H_5NH_2$** by ammonolysis

$C_2H_5NH_2 + HNO_2$ → **$C_2H_5OH + N_2 + H_2O$**

Primary aliphatic amines react with nitrous acid to give alcohols with evolution of N_2.

Q86 Text Solution:

Arrhenius equation:

$\ln\ k = \ln\ A - \frac{E_a}{RT}$

Given:

$\ln k = 14.34 - \frac{1.25\times10^4}{T}$

So,

$\frac{E_a}{R} = 1.25 \times 10^4$

$E_a = 1.25 \times 10^4 \times 1.987 = 24837.5$ cal mol^{-1}

$= 24.84\ \text{kcal mol}^{-1}$

Q87 **Text Solution:**

$K_p = K_c(RT)^{\Delta n}$

For $K_p < K_c$, Δnshould be negative.

For:

$N_2(g) + 3H_2(g) \rightleftharpoons 2\,NH_3(g)$

$\Delta n = 2 - (1 + 3) = -2$

So,

$\Delta n = 2 - (1 + 3) = -2$

Q88 **Text Solution:**

Among Mg, Mg^{2+}, Al, Al^{3+} :

$\text{Largest} = Mg$

$\text{Smallest} = Al^{3+}$

Reason: atomic size decreases from Mg to Al across the period, and cations are smaller than neutral atoms. Also, Mg^{2+} and Al^{3+} are isoelectronic, but Al^{3+} has higher nuclear charge, so it is smaller.

Q89 **Text Solution:**

Chloroform and acetone show **negative deviation** from Raoult's law because new **stronger intermolecular attraction** forms between them.

$Cl_3\,CH\ \ldots\ \ O = C(CH_3)_2$

This is hydrogen bonding between the acidic H of chloroform and oxygen of acetone.

Q90 **Text Solution:**

With **anhyd. $AlCl_3$, dark/cold**, benzene undergoes electrophilic substitution with excess chlorine:

$C_6H_6 + 6\,Cl_2 \rightarrow C_6\,Cl_6 + 6\,HCl$

So, **X = hexachlorobenzene**, has **6 Cl atoms**.

With **UV, 500 K**, benzene undergoes addition:

$C_6H_6 + 3\,Cl_2 \rightarrow C_6H_6\,Cl_6$

So, **Y = benzene hexachloride/BHC**, has **6 Cl atoms**.

Q91 **Text Solution:**

Root hairs arise from the **epidermal cells** in the **region of maturation** of the root. They help in absorption of water and minerals.

Q92 **Text Solution:**

In **gymnosperms**, ovules are **not enclosed by ovary wall**, so seeds remain exposed/naked. **Pinus** is a gymnosperm.

Q93 **Text Solution:**

In **lac operon**:

z gene → β-galactosidase, **y gene → permease**, **a gene → transacetylase**, and **i gene → repressor**.

Q94 **Text Solution:**

Bioprospecting is defined as exploring molecular, genetic and species-level diversity for products of economic importance.

Q95 **Text Solution:**

- **Genetically modified organism** → **Bt cotton** → **II**
- **Thermostable DNA polymerase** → ***Thermus aquaticus*** → **III**
- **Ti plasmid** → ***Agrobacterium tumefaciens*** → **I**
- **pBR322** → cloning vector of ***Escherichia coli*** → **IV**

Q96 **Text Solution:**

- **Productivity** → Rate of biomass production → **III**
- **Net primary productivity (NPP)** → GPP – respiration losses → **I**
- **Gross primary productivity (GPP)** → Rate of production of organic matter during photosynthesis → **IV**
- **Secondary productivity** → Rate of formation of new organic matter by consumers → **II**

Q97 **Text Solution:**

The ongoing sixth extinction is different because its rate is 100–1000 times faster than pre-human times, mainly due to human activities.

Q98 Text Solution:

α-helix and **β-pleated sheet** are forms of **secondary structure** of proteins.

Q99 Text Solution:

In grasses, **bulliform cells** become flaccid during water stress and make the leaves **curl inward**, reducing water loss.

Q100 Text Solution:

1. **Denaturation**: DNA strands separate
2. **Annealing**: Primers bind
3. **Extension**: Taq polymerase extends primers

Q101 Text Solution:

- **G_1 phase** → Cell grows, metabolically active, **DNA not replicated** → **II**
- **S phase** → **DNA synthesis**, DNA amount doubles → **III**
- **G_2 phase** → Proteins synthesized, growth continues → **IV**
- **M phase** → Actual cell division occurs → **I**

Q102 Text Solution:

- **A correct:** Amazon rainforest cleared for soya beans = **habitat loss**.
- **B correct:** Steller's sea cow and passenger pigeon became extinct due to **over-exploitation**.
- **C incorrect:** Nile perch caused extinction/decline of many **cichlid fishes**, not growth.
- **D correct:** Water hyacinth is an **invasive alien species**.
- **E incorrect:** In co-extinction, associated plant/animal species **are affected** and may also become extinct.

Q103 Text Solution:

All statements are **correct**.

A transcription unit has:

- **Promoter**
- **Structural gene**
- **Terminator**

Promoter is near the **5′ end** of structural gene, binds **RNA polymerase**, and helps decide **template/coding strand**. Terminator is near the **3′ end of coding strand** and ends transcription.

Q104 Text Solution:

Statement (3) is not true because the first word represents genus, and the second word represents specific epithet.

Example: *Mangifera indica* → *Mangifera* = genus, *indica* = specific epithet.

Q105 Text Solution:

- **Decomposition** → Breaking down complex organic matter into inorganic substances → **III**
- **Detritus** → Dead remains of plants and animals including fecal matter → **IV**
- **Mineralisation** → Release of inorganic nutrients by microbes → **II**
- **Humification** → Accumulation of dark coloured amorphous colloidal substance → **I**

Q106 Text Solution:

Nucleolus is the site of **active rRNA synthesis** and ribosomal subunit assembly.

So, for ribosomal RNA synthesis → **Nucleolus**.

Q107 Text Solution:

RQ = **CO_2 evolved / O_2 consumed**

= **102 / 145 = 0.70 approx.**

So RQ is **between 0.5 and 0.95**. This indicates a **fat/lipid-like substrate (Tripalmitin)**.

Q108 Text Solution:

- **Incomplete dominance** → Flower colour in *Antirrhinum* → **II**
- **Co-dominance** → ABO blood groups → **IV**
- **Pleiotropy** → Phenylketonuria in humans → **III**
- **Polygenic inheritance** → Human skin colour → **I**

Q109 Text Solution:

Correct DNA fingerprinting sequence:

A Isolation of DNA and digestion by restriction endonucleases

→ **E** Separation of DNA fragments by electrophoresis
→ **C** Transfer to synthetic membrane
→ **B** Hybridisation with labelled VNTR probe
→ **D** Detection by autoradiography

Q110 Text Solution:

A ✅ Histone octamer = **8 histone molecules**
B ❌ Histones are **positively charged basic proteins**, not negatively charged.
C ✅ Rich in **lysine and arginine**
D ❌ **Negatively charged DNA** wraps around positively charged histone octamer.
E ✅ Higher packaging needs **Non-histone Chromosomal (NHC) proteins**.

Q111 Text Solution:

Incorrect statements:

- **A incorrect:** Water-splitting complex is associated with **PS II**, not PS I.
- **D incorrect: C_4 plants** exhibit Kranz anatomy, not C_3 plants.

Correct statements:

- **B correct:** C_4 plants ultimately use the **C_3/Calvin pathway** as main biosynthetic pathway.
- **C correct:** Photorespiration is essentially absent/negligible in C_4 plants.
- **E correct:** ATP synthesis in chloroplast occurs by **chemiosmosis**.

Q112 Text Solution:

Correct sequence:
Isolation of single cells → Digestion of cell walls → Isolation of naked protoplasts → Fusion of protoplasts → Growing hybrid protoplast into new plant
So, **D → A → B → C → E**.

Q113 Text Solution:

- **Conjunctive tissue** → Tissue between xylem and phloem → **III**
- **Casparian strips** → Endodermal cells with suberin deposition → **IV**
- **Subsidiary cells** → Specialised cells near guard cells → **I**
- **Starch sheath** → Endodermal cells rich in starch → **II**

Q114 Text Solution:

In the **phase of elongation**, cells show:

- **Cell enlargement**
- **Increased vacuolation**
- **New cell wall deposition**

Large conspicuous nuclei are mainly characteristic of **meristematic cells**, not elongation phase.

Q115 Text Solution:

A. **2,4-D → Herbicide**
B. **GA_3 → Brewing industry**
C. **Kinetin → Nutrient mobilisation**
D. **ABA → Stimulation of stomatal closure**

Q116 Text Solution:

In the Calvin cycle, CO_2 fixation/carboxylation is catalysed by RuBisCO.

Q117 Text Solution:

For **1 glucose molecule**, Calvin cycle fixes **6 CO_2**.
Requirement per CO_2 = **3 ATP + 2 NADPH**
So for 6 CO_2:
ATP = 6 × 3 = 18
NADPH = 6 × 2 = 12
Hence, **18 ATP and 12 NADPH**.

Q118 Text Solution:

Floral formula of Solanaceae is:

$$\oplus\ ⚥\ K_{(5)}\ \overset{\frown}{C_{(5)}\ A_5}\ \underline{G}_{(2)}$$

It has **actinomorphic, bisexual flower; united sepals, united petals, 5 epipetalous stamens, bicarpellary syncarpous superior ovary**.

Q119 Text Solution:

In situ conservation = conservation in the **natural habitat**.
NCERT lists **sacred groves** under in situ conservation.

Wildlife safari parks, botanical gardens, and seed banks are **ex situ** methods.

Q120 Text Solution:

- **A true:** Restriction endonucleases are called **molecular scissors**.
- **B true:** They restrict growth of bacteriophages in bacteria like *E. coli.*
- **C not true:** They do **not always cut only at the centre** of palindromic sites; cuts may be staggered or blunt.
- **D not true:** Removing nucleotides from DNA ends is done by **exonucleases**, not endonucleases.
- **E true:** They recognise specific **palindromic sequences**.

Q121 Text Solution:

In **racemose inflorescence**, main axis keeps growing and flowers are arranged **acropetally** — older flowers at the base, younger flowers towards the apex.

Q122 Text Solution:

Correct sequence of microsporogenesis:
Sporogenous tissue → Pollen mother cells → Microspore tetrads → Pollen grains

Q123 Text Solution:

- **A incorrect:** Lipids are generally **water-insoluble**.
- **B correct:** Proteins are **polypeptides**.
- **C correct:** Polysaccharides are **long chains of sugars**.
- **D incorrect:** Adenine and guanine are **purines**, not substituted pyrimidines.
- **E correct:** Almost all enzymes are **proteins**.

Q124 Text Solution:

- **A true:** Fertilised egg → **female** honeybee (queen/worker).
- **B true:** Unfertilised egg → **male** by **parthenogenesis**.
- **C true:** Male is **haploid**, female is **diploid**.
- **D false:** Males produce sperms by **mitosis**, not meiosis.
- **E true:** Honeybee shows **haplodiploid sex determination**.

Q125 Text Solution:

Heterophylly means different leaf forms in response to environment. This ability of plants to alter development according to environmental conditions is called **plasticity**.

Q126 Text Solution:

- **A correct:** Amino acids are **substituted methanes**.
- **B incorrect:** Serine is **not aromatic**; it is a neutral amino acid.
- **C correct:** Valine is a **neutral amino acid**.
- **D incorrect:** Lysine is **basic**, not acidic.

Q127 Text Solution:

The Evil Quartet of biodiversity loss includes:

1. **Habitat loss and fragmentation**
2. **Over-exploitation**
3. **Alien species invasions**
4. **Co-extinctions**

Q128 Text Solution:

A. **Glycolysis → Cytoplasm**
B. **ETS → Inner mitochondrial membrane**
C. **Accumulation of protons → Intermembrane space**
D. **Krebs' cycle → Mitochondrial matrix**

Q129 Text Solution:

After **triple fusion**, the central cell becomes the **primary endosperm cell (PEC)**, which is **triploid (3n)**.
Zygote is **2n**, synergid is **n**.

Q130 Text Solution:

Xenogamy = transfer of pollen grains from anther to stigma of a different plant of the same species. So it brings genetically different pollen grains to the stigma.

Q131 Text Solution:

- **Marginal placentation → Pea**
- **Axile placentation → Lemon**
- **Parietal placentation → Mustard**

- **Basal placentation → Marigold**

Q132 Text Solution:

NCERT: Whittaker's Five Kingdom classification criteria include **cell structure, body organisation, mode of nutrition, reproduction, and phylogenetic relationships**.

Presence of flagellum is **not** a main criterion.

So, **C is excluded**.

Q133 Text Solution:

A. **Trypsin → Enzyme**

B. **Morphine → Alkaloid**

C. **Concanavalin A → Lectin**

D. **Collagen → Intercellular ground substance**

Q134 Text Solution:

- **A correct:** DNA is cut by restriction enzymes, called **molecular scissors**.
- **B correct:** DNA fragments separate by **size** in agarose gel electrophoresis.
- **C incorrect:** DNA fragments are **not visible without staining**.
- **D incorrect:** Ethidium bromide-stained DNA is seen under **UV light**, not visible light.

Q135 Text Solution:

In sickle-cell anaemia, **glutamic acid is replaced by valine** at the **6th position of the β-globin chain** of haemoglobin.

Q136 Text Solution:

Cortisol → anti-inflammatory reactions.

Aldosterone → Na^+ and water reabsorption from renal tubules.

Cholecystokinin → pancreatic enzymes and bile juice secretion.

Progesterone → formation of alveoli in mammary glands.

Q137 Text Solution:

Correct order:

Fall in GFR → Renin converts Angiotensinogen into Angiotensin I, followed by Angiotensin II → Vasoconstriction by Angiotensin II and release of Aldosterone → Reabsorption of Na^+ and water from distal parts of tubule due to Aldosterone → Increase in blood pressure and Glomerular filtration rate.

Q138 Text Solution:

1. Pulmonary ventilation (C) Air enters lungs (O_2 in) and CO_2 is expelled. This is the first step.

2. Diffusion across alveolar membrane (B) O_2 diffuses from alveoli → blood CO_2 diffuses from blood → alveoli.

3. Transport of gases by blood (E) O_2 is carried by haemoglobin to tissues CO_2 is transported back to lungs.

4. Diffusion between blood and tissues (A) O_2 moves from blood → tissues CO_2 moves from tissues → blood.

5. Cellular respiration (D) Cells use O_2 to produce energy (ATP) and release CO_2.

Q139 Text Solution:

Female *Anopheles* injects **sporozoites** → they reach **liver** → multiply asexually in liver cells → enter and multiply in **RBCs** → some develop into **gametocytes**.

Q140 Text Solution:

In **pBR322**, the **BamHI site lies within the tetracycline resistance gene (tet^R)**.

So insertion of foreign DNA at **BamHI** causes **insertional inactivation of tet^R**, leading to loss of **tetracycline resistance**.

Q141 Text Solution:

Lyases catalyse removal of groups from substrates by mechanisms other than hydrolysis, often forming a **double bond (C=C)**.

Q142 Text Solution:

Neurotransmitters released from the pre-synaptic knob bind to **specific receptors on the post-synaptic membrane** to transmit the impulse.

Q143 Text Solution:

Mother heterozygous A = **I^Ai**

Father heterozygous B = **I^Bi**

Cross: **I^Ai × I^Bi** → offspring:

I^AI^B, I^Ai, I^Bi, ii

So **O blood group (ii)** probability = **1/4 = 25%**.

Q144 Text Solution:

A. **ERV** → 1000–1100 mL
B. **RV** → 1100–1200 mL
C. **IRV** → 2500–3000 mL
D. **TV** → 500 mL

Q145 Text Solution:

Whale and bat forelimbs are **homologous organs** with common origin but different functions, so they show **divergent evolution**, not convergent evolution.

Q146 Text Solution:

Male frogs have **vocal sacs** and a **copulatory pad on the first digit of forelimbs.**
Bulging eyes, webbed feet, and olive-green skin are common frog features, not male-specific.

Q147 Text Solution:

These features match **Cyclostomata**:
ectoparasitic on fishes, **circular sucking mouth**, **6–15 pairs of gill slits**, no scales and no paired fins.
Example: ***Petromyzon*** **(Lamprey)**.

Q148 Text Solution:

About **65 mya** dinosaurs disappeared.
About **500 mya** invertebrates became active.
About **350 mya** jawless fishes evolved.
About **320 mya** seaweeds and a few plants existed.

Q149 Text Solution:

A. **Progestasert** → Hormone-releasing IUD
B. **Multiload 375** → Copper-releasing IUD
C. **Diaphragm** → Rubber barrier used by females
D. **Saheli** → Oral contraceptive

Q150 Text Solution:

Eosinophils are about **2–3%** of WBCs → 2–3% of 8000 = **160–240**.
Lymphocytes are about **20–25%** → 20–25% of 8000 = **1600–2000**.

Q151 Text Solution:

Nicotine	stimulates adrenal gland to release catecholamines
Morphine	effective sedative and painkiller
Heroin	depressant; slows down body functions
Cocaine	causes euphoria and increased energy

Q152 Text Solution:

Transgenic animals can produce useful biological products, e.g. **α-1-antitrypsin**, used to treat **emphysema**.

Q153 Text Solution:

All three are **bony fishes** of class **Osteichthyes**:
Flying fish = *Exocoetus*, Fighting fish = *Betta*, Angel fish = *Pterophyllum*.

Q154 Text Solution:

Adult echinoderms are **radially symmetrical**; larvae are bilateral.
Reptiles are **poikilothermic/cold-blooded**, so they do not maintain constant body temperature.

Q155 Text Solution:

Ribosomes are **non-membrane-bound organelles** present in both **prokaryotic and eukaryotic cells.**
Mitochondria and lysosomes are membrane-bound; centrosome is mainly in animal cells.

Q156 Text Solution:

$$\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$$

Reason:
Verhulst-Pearl logistic growth equation includes **carrying capacity (K)** and shows growth slowing as population size **N** approaches **K**.

Q157 Text Solution:

A is incorrect: Erythroblastosis foetalis occurs when **mother is Rh^- and foetus is Rh^+**.
E is incorrect: Anti-Rh antibodies are given **immediately after the first delivery**, not second.

Q158 Text Solution:

Streptokinase removes clots from blood vessels. Statins lower blood cholesterol. Lipases are used in detergent formulations. Cyclosporin A is an immunosuppressive agent.

Q159 Text Solution:

Molluscs → branchial respiration.

Reptiles → pulmonary respiration only.

Adult amphibians → pulmonary + cutaneous respiration.

Amoeba → cellular respiration.

Q160 Text Solution:

In sickle-cell anaemia, the 6th codon of β-globin changes from **GAG → GUG**, causing **Glu → Val** substitution and Hb polymerisation.

Q161 Text Solution:

C is false because Ca^{2+} **activates actin sites** by removing masking, it does not inactivate actin.

A, B, D and E correctly describe excitation and sliding filament mechanism.

Q162 Text Solution:

A is false: Human skull is **dicondylic**, not monocondylic.

D is false: There are 12 pairs of ribs. Each rib is a thin flat bone connected dorsally to the vertebral column and ventrally to the sternum. It has two articulation surfaces on its dorsal end and is hence called bicephalic.

B true: Joints between adjacent vertebrae are **cartilaginous joints**.

C true: Humans have **7 cervical vertebrae**.

E true: **Occipital condyles** articulate with the **atlas vertebra**.

Q163 Text Solution:

A ❌ Spermatogonia divide via **mitosis** (equational division) to maintain their population, not directly via meiosis.

B ❌ Primary spermatocytes undergo **meiosis I**, not mitosis.

C ✅ Secondary spermatocytes undergo **meiosis II** to form haploid spermatids.

D ❌ Spermatids do **not** form spermatozoa by mitosis.

E ✅ Spermatids transform into spermatozoa by **spermiogenesis**.

Q164 Text Solution:

JGA is formed by cellular modifications in the **DCT** and the **afferent arteriole** at their contact point.

Q165 Text Solution:

Ophrys orchid shows **sexual deceit** by mimicking the female bee/wasp, attracting the male insect for **pseudocopulation**, which helps in pollination.

Q166 Text Solution:

A true: Hepatic portal system connects **intestine and liver**.

C true: In female frog, **ureters and oviducts open separately into cloaca**.

E true: **Sinus venosus opens into right atrium**.

B false: Frog has **10 pairs** of cranial nerves, not 12.

D false: **Optic lobes are midbrain**, not hindbrain.

Q167 Text Solution:

A.The foetus movement starts and hair appears on the head→ **20 weeks of pregnancy**

B.The foetus develops limbs and digits → **8 weeks of pregnancy**

C. The foetus develops external genital organs → **12 weeks of pregnancy**

D. The foetus body is covered with fine hair; eyelids separate and eyelashes are formed → **24 weeks of pregnancy**

Q168 Text Solution:

In grasshopper, **XO type sex determination** occurs:

Male = XO = 23 chromosomes, Female = XX = 24 chromosomes.

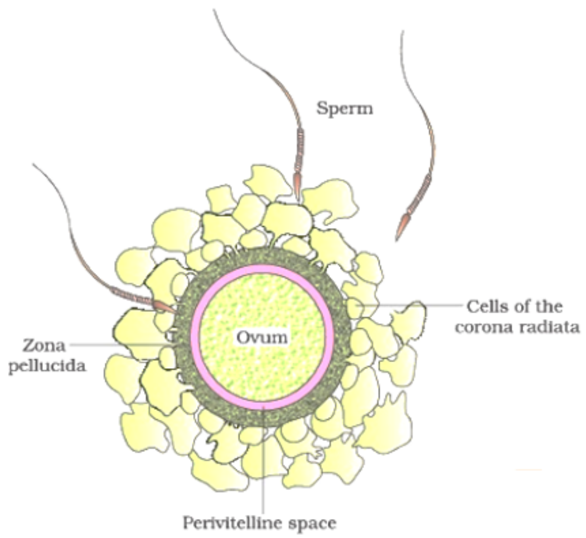
Q169 Text Solution:

Male honeybees are **haploid** and produce sperms by **mitosis**, not meiosis. This is due to haplodiploid sex determination.

Q170 Text Solution:

From outer to inner around ovum:

Corona radiata → Zona pellucida → Perivitelline space → Plasma membrane of ovum.



Q171 Text Solution:

Large holes in Swiss cheese are due to CO_2 production by *Propionibacterium sharmanii* during fermentation.

Q172 Text Solution:

cryIAc controls cotton bollworms, while **cryIAb** controls corn borer.

Q173 Text Solution:

The pyramid of biomass in sea is generally inverted because the biomass of fishes far exceeds that of phytoplankton.
Energy pyramid is **always upright**.

Q174 Text Solution:

Correct option: (2)

GIFT, which stands for **Gamete Intrafallopian Transfer**, is a specific assisted reproductive technique designed for women who cannot produce their own eggs but possess a healthy reproductive environment.
Based on the definition provided:

- **The Procedure**: It involves the transfer of an **ovum** (egg) collected from a donor directly into the **fallopian tube** of another female.
- **The Recipient**: This technique is used for a female who cannot produce her own ovum but can provide a **suitable environment** for fertilization and subsequent development.

Unlike standard IVF where fertilization happens in a laboratory, in GIFT, the fertilization takes place **naturally** inside the recipient's fallopian tube.
Option (4) is **ZIFT**, not GIFT.

Q175 Text Solution:

A is false because mitochondria are **not** part of the endomembrane system.
E is false because mitochondria are **double membrane-bound**, not single membrane-bound.

Q176 Text Solution:

A true: Human RBC membrane has about **52% protein**.
B true: Phospholipids are arranged in a **bilayer**.
D true: Lipid tails are **hydrophobic** and face inward.
C is false: **Mesosomes occur in prokaryotes**, not eukaryotic cells.
E is false: **Glycocalyx is characteristic of bacterial cell envelope**, not eukaryotic plasma membrane in NCERT context.

Q177 Text Solution:

Tetany	wild contraction of muscles due to low Ca^{2+}
Arthritis	inflammation of joints
Myasthenia gravis	autoimmune disorder affecting neuromuscular junction
Muscular dystrophy	progressive degeneration of skeletal muscles

Q178 Text Solution:

Correct sequence of appearance:
Ramapithecus → Homo habilis → Homo erectus → Neanderthal → Homo sapiens
This follows the general NCERT human evolution order from ape-like ancestor to modern human.

Q179 Text Solution:

Aptenodytes = **Penguin**, a flightless bird whose forelimbs are modified into **paddle-like flippers** for swimming.

Q180 Text Solution:

Mutualism benefits both species.
In parasitism, parasite benefits and host is

harmed.
In amensalism, one species is harmed and the other is unaffected.

A is false because in **commensalism**, one benefits and the other is **unaffected**, not harmed.